

A-MES

AUTOMOTIVE MES · SEOYON E-HWA

VOL 04

Production Planning (PP)

Forecast · MRP · Work Order · Calendar · OEE

Document · L3 Detailed Design Specification

Generated · 05-26-2026 · Build 20260526-175635

Reference · Plant A " SAV Detroit, MI | Plant B " GEO Birmingham, AL

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A-MES

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WH Warehouse

PP Prod. Plan

INJ Injection

IMG Wrapping

PNT Painting

QC Quality

FG Fin. Goods

VOL.4 · PP MODULE

PRODUCTION

생산계획

OVERVIEW · 개요

FLOW

Process Flow

전체 흐름

KPI

Module KPI

핵심 지표

WO

WO Status

WO 상태 흐름

DEMAND · 수요

PP-01

Forecast

Forecast 관리

PP-02

SAP B1 Import

공급계획 수신

PLAN · 계획

PP-03

Plan & Confirm

생산계획 수립

PP-04

Work Order

WO 관리

MATERIAL · 자재

PP-05

MRP

자재 소요 계획

PP-06

Purchase Req.

자재발주 요청

RELEASE · 발행

PP-07

WO Release

WO 발행

PP-CAL

Calendar

생산 캘린더

PP-OTD

Delivery KPI

납기 현황

VOL.4 · PP · PRODUCTION PLANNING · 생산 계획

PRODUCTION PLANNING

생산 계획 — Forecast · SAP B1 Sync · MRP · Auto Purchase Req · WO Release

Full production planning cycle from customer demand intake to shop-floor release. Pipeline: customer Forecast (SCM portal) → SAP B1 supply plan import → MES production plan → MRP material requirements → Purchase Requisition (SAP B1 Service Layer API) → Work Order release → POP shop-floor (INJ/IMG/PNT). Dual MRP: A-MES for real-time WO blocking, SAP B1 for mid/long-term procurement.

고객 수요 수신부터 현장 발행까지 생산계획 전 사이클. 파이프라인: 고객 Forecast (SCM 포탈) → SAP B1 공급계획 Import → MES 생산계획 수립 → MRP 자재 소요 → 자재발주 요청서 (SAP B1 Service Layer API) → WO 발행 → POP 현장 (INJ/IMG/PNT). 이중 MRP: A-MES는 실시간 WO 차단, SAP B1은 중 장기 발주 담당.

SCREENS13

ERP
SAP
B1

CUSTOMERS
SAV ·
GEO

MRP
Dual
(MES+B1)

PR
INTERFACE
Service
Layer
API

ROL
A ·

Click any screen card / icon / pill for L3 detailed spec · 각 화면 카드 클릭 시 L3 상세 설계서

PP-01 Forecast

PP-02 SAP B1 Import

PP-03 Plan & Confirm

PP-04 Work Order

PP-05 MRP

PP-06 Purchase Req.

PP-07 WO Release

PP-CAL Calendar

PP-OTD Delivery KPI

PP-LSB Line Schedule Board

PP-ODM Downtime Monitor

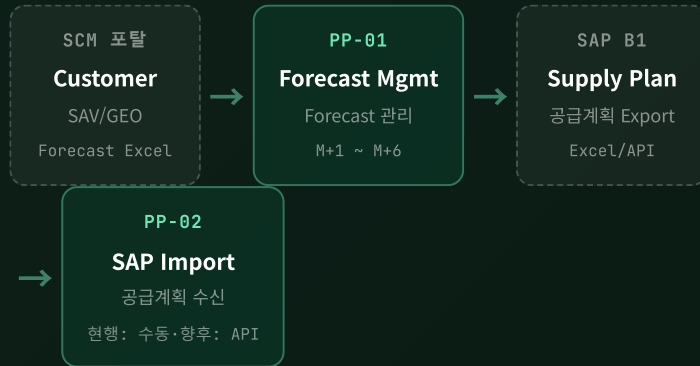
PP-OEE Line OEE

PP-DTL Downtime Log

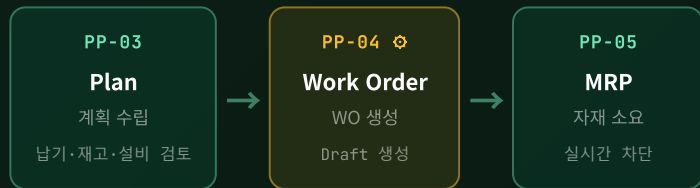
END-TO-END FLOW · 전체 업무 흐름

▼ DEMAND → PLAN → MATERIAL → RELEASE

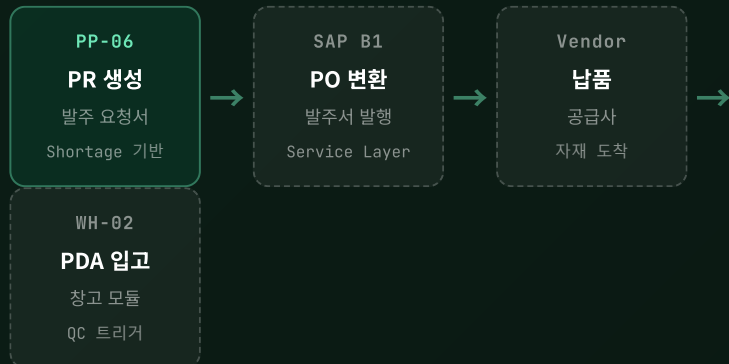
① DEMAND INTAKE · 수요 수신



② PLAN & WO CREATION · 계획·WO 생성



③ MATERIAL PROCUREMENT · 자재 발주

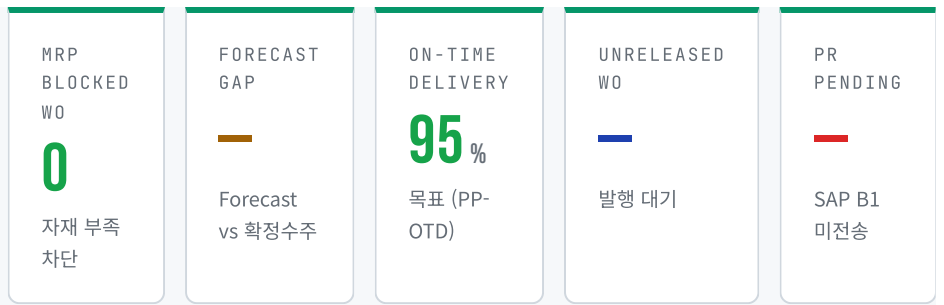


④ RELEASE TO SHOP FLOOR · 현장 발행

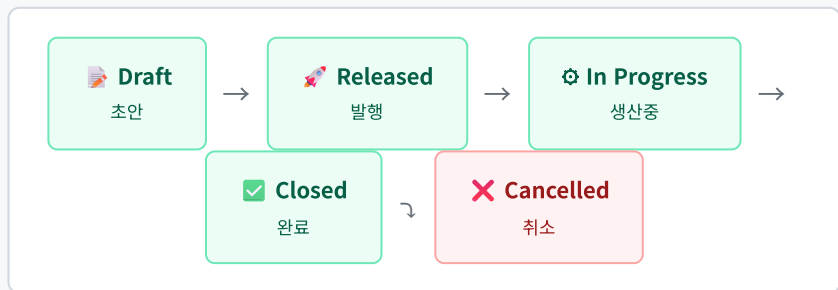


⚙️ Core - 핵심 게이트 PP 화면 외부 모듈/시스템

▼ MODULE KPI · 핵심 지표



▼ WO STATUS FLOW · WO 상태 흐름



▼ 9 SCREENS · 화면 설계

PP-01

Customer
Forecast
Management

Web PC

1st Dev

L3 DETAIL →

고객사 Forecast 수
신 및 관리

What you'll see: 고객 SCM 포탈 Forecast Excel 다운로드
+ A-MES Import + 품번 매핑 화면

Forecast 대비 확정 수주 편차 자동 추적 (Gap > 20% 시
PLAN/MGR 알림)

INPUT

SCM Excel
(SAV/GEO)

OUTPUT

Forecast
Table + Gap
KPI

HORIZON

M+1 ~ M+6

ALERT

Gap > 20%

EN · DESCRIPTION

Receives and manages
customer demand forecasts
(M+1 to M+6) from SAV/GEO
SCM Portal. Downloads Forecast
Excel and imports into A-MES
with item mapping. Tracks gap
between forecast and confirmed

KO · 설명

고객사(SAV/GEO) SCM 포탈에서
Forecast Excel을 다운로드하여 A-
MES에 Import. 품번 매핑 후
Forecast 대비 확정 수주 편차 추적.
Forecast 재업로드 시 이력 보관 후
UPDATE. 고객사 품번 미매핑 시 경
고.

orders. Forecast re-upload
retains history. Unmapped
customer item number triggers
warning.



Manual Import

고객사 포탈에서 Excel 다운로드 후 업로드.
매월 정기



Item Mapping

고객 품번 ↔ MD-01 자사 품번 매핑.
미매핑 경고 (BR-F001)



Gap Tracking

Forecast vs 확정 수주 편차 자동 계산.
>20% 알림 (BR-F002)



History Retention

재업로드 시 이력 보관 후 UPDATE.
PP_ForecastHistory (BR-F003)

관련 BR ▶

BR-PP-F001

BR-PP-F002

BR-PP-F003

PP-02



SAP B1 Supply Plan Import

SAP B1 공급계획 수신

Web PC

SAP B1

L3 DETAIL →

What you'll see: SAP B1 확정 공급계획(판매오더) Import



화면 — 현행 Excel 수동 / 향후 Service Layer API 자동

PP-03 생산계획 수립의 입력 데이터



INPUT

SAP B1
Excel / API



OUTPUT

Supply
Plan Table
→ PP-03



CYCLE

현행 일 1회 ·
향후 1h



INTEGRATION

Service
Layer
(planned)

EN · DESCRIPTION

KO · 설명

Receives confirmed supply plans (sales orders) from SAP Business One. **Current:** SAP B1 Excel Export → A-MES manual Import. **Future roadmap:** SAP B1 Service Layer REST API auto-polling every 1 hour. Drives PP-03 production planning.

SAP Business One의 확정 공급계획(판매오더)을 A-MES로 수신. **현행:** SAP B1 Excel Export → A-MES 수동 Import. **향후:** SAP B1 Service Layer REST API 자동 폴링(1시간)으로 전환 예정. PP-03 생산계획 수립 입력.



Current: Excel

SAP B1 Export Excel → 수동 Import.
현 운영 방식



Future: API Poll

Service Layer REST API 자동 폴링.
1시간 간격 예정



Diff Detect

이전 Import 대비 변경 라인 강조.
PLAN 검토 효율



→ PP-03

확정 시 PP-03 생산계획으로 인계.
자동 분기

관련 BR ▶ 참조: PP-03 plan rules

PP-03



Production Planning & Confirmation

생산계획 수립 및 확인

Web PC

1st Dev

L3 DETAIL →



What you'll see: PP-02 수신 공급계획 + FG 재고 + 설비 부하 + Forecast 비교 화면 — 4 변수 한 번에 검토
PLAN 담당자가 생산 가능 여부 확정 → WO 일괄 생성 대상 마킹



INPUT

PP-02 Plan
+ FG +



OUTPUT

Confirmed
Plan → PP-



ROLE

PLAN
Manager



UI

4-Variable
Compare

Equip

04

EN · DESCRIPTION

PLAN manager reviews received supply plans against current FG inventory, equipment load, and Forecast reference. Confirms feasible items for batch WO creation in PP-04. Out-of-capacity items are flagged with overflow indicator for re-scheduling.

KO · 설명

PLAN 담당자가 PP-02 공급계획을 완제품 재고·설비 부하·Forecast 참고치와 대조하여 생산 가능 여부를 검토하고 PP-04 WO 일괄 생성 대상을 확정. 캐파 초과 품목은 overflow 표시로 재배정 권유.



FG Stock Check

현 FG 재고 vs 출하 요구.
잉여 자동 차감



Capacity Load

설비/라인 가용 캐파 vs 계획 부하.
초과 빨강 표시



Forecast Reference

PP-01 Forecast 참고 표시.
트렌드 검증



→ PP-04 Batch

확정 행 일괄 WO 생성 트리거.
Ready 토글

관련 BR ▶ 참조: PP-04 (WO 생성), PP-05 (MRP)

PP-04 · ⚙️
CORE



Work Order Management

작업지시 (WO) 관리

[Web PC](#)[Core](#)[L3 DETAIL →](#)

What you'll see: WO 4-상태 (Draft → Released → In Progress → Closed) 그리드 + 라우팅 A/B 배지 + D-Day + 진행률 + SAP B1 링크



Phase-0 미완료 품목은 자동 차단 (BR-PP-001)

INPUT

PP-03
Confirmed

OUTPUT

WO Draft →
MRP/Release

STATES

Draft·Released·InProg·Closed

EN · DESCRIPTION

Manages Work Orders from Draft through Released, In Progress, to Closed. Displays routing type badge (A/B), progress bar, D-Day indicator, and SAP B1 supply plan link. Phase-0 incomplete items are blocked (BR-PP-001). Reserved qty auto-released on WO cancel.

KO · 설명

WO를 Draft → Released → In Progress → Closed 상태로 관리. 라우팅 A/B 배지·진행률·납기 D-Day·SAP B1 공급계획 연결 표시. Phase-0 미완료 품목은 자동 차단 (BR-PP-001). 취소 시 예약 수량 자동 해제.



Routing A/B Badge

감싸기(A) / 도색(B) 시각 구분.
MD-04 참조



Progress Bar

실시간 진행률 (POP 입력 연동).
INJ/IMG/PNT



D-Day Indicator

납기 잔여일 + D-3 강조(BR-008).
캘린더 빨강



Phase-0 Gate

마스터 미완료 차단 (BR-001).
Item/BOM/BOP 필수

관련 BR ▶

BR-PP-001

BR-PP-004

BR-PP-008

PP-05

MRP —
Material



Web PC

Batch

L3 DETAIL →

Requirements

Planning

자재 소요 계획 — 이
중 MRP (MES + SAP
B1)

**What you'll see: BOM 기반 자재 소요량 계산 결과 + WO별
부족 자재 리스트 + 자동 차단 표시 + 야간 배치 이력**



A-MES MRP: 실시간 WO 차단 / SAP B1 MRP: 중장기 발주 (역할 분
리)



INPUT

**WO + BOM
+ WH Stock**



OUTPUT

**Shortage
Qty → PP-
06**



SCHEDULE

**매일 00:00 +
On-
Demand**



BLOCK

**Shortage >
0 → WO 차
단**

EN · DESCRIPTION

Runs MRP to calculate material shortage quantities. **A-MES MRP**: real-time WO blocking, immediate availability check. **SAP B1 MRP**: mid/long-term procurement planning, PO generation. Dual MRP runs in parallel with clear role separation. Nightly batch at 00:00 recomputes all shortages.

KO · 설명

MRP를 실행하여 자재 부족 수량 계산. **A-MES MRP**: 실시간 WO 차단, 즉시 가용성 확인. **SAP B1 MRP**: 중장기 자재 소요·공급업체 발주 담당. 두 MRP 역할 분리하여 병행 운영. 매일 00:00 배치 전체 재계산.



A-MES Realtime

WO 생성 즉시 ShortageQty 계산.
BR-PP-002 차단



B1 Mid-Term

SAP B1이 중장기 발주 계획.
PO 생성



Nightly Batch

매일 00:00 전체 재계산.
BR-PP-005



→ PP-06 PR

Shortage → PR 자동 생성 대상.

자동 분기

관련 BR ▶

BR-PP-002

BR-PP-005

PP-06



Purchase Requisition → SAP B1

Web PC

SAP B1

L3 DETAIL →

자재발주 요청서 (PR)
→ SAP B1 전송

What you'll see: MRP 부족 자재 기반 PR 생성 폼 + SAP B1 전송 버튼 + 전송 결과 DocNum 수신 + 실패 시 재시도
Service Layer API (POST /PurchaseRequests) · 실패 시 3회 재시도

INPUT

MRP Shortage

TARGET

SAP B1 PR (DocNum)

API

Service Layer REST

RETRY

3회 자동

EN · DESCRIPTION

Generates Purchase Requisition records from MRP shortage results and transmits them to SAP B1 via Service Layer API (POST /PurchaseRequests). Tracks SAP B1 DocNum after successful transmission. 3 auto-retries on failure; persistent failure escalates to PLAN team.

KO · 설명

MRP 부족 자재를 기반으로 발주요청서(PR)를 생성하고, SAP B1 Service Layer API(POST /PurchaseRequests)로 전송. 전송 성공 시 SAP B1 DocNum 수신·저장. 실패 시 3회 자동 재시도; 지속 실패 시 PLAN 팀 알림.



Auto Generate

MRP Shortage → PR row 자동 생성.
수량·납기 자동



SAP B1 API

Service Layer POST 전송.
DocNum 수신



Retry ×3

전송 실패 시 자동 재시도.



Failure Alert

3회 모두 실패 시 PLAN 팀 알림.
이메일 + Slack

관련 BR ▶

BR-PP-006

PP-07 · ⚙️
CORE



Work
Order
Release

WO 발행 및 생산
착수 — 4중 검증
게이트

Web PC

Phase 0 + 1st

L3 DETAIL →



What you'll see: WO Released 전환 전 4중 조건 검증 화면
(Phase-0·MRP·설비·납기) + 통과 시 자동 자재 예약 + POP
노출

발행 핵심 게이트 — 한 가지라도 실패하면 차단

INPUT

WO Draft

OUTPUT

WO
Released
→ POP

CHECKS

4
(P0·MRP·Equip·
납기)

AUTO

자재 예약 +
자동 노출

EN · DESCRIPTION

Validates all release conditions before transitioning WO to Released status: **(1) Phase-0 complete, (2) MRP no shortage, (3) equipment available, (4) delivery lead time sufficient.** All pass → WO release → auto material reservation → POP screen exposure (INJ/IMG/PNT).

KO · 설명

WO Released 전환 전 4가지 조건 순서대로 검증: **(1) Phase-0 완료, (2) MRP 자재 충족, (3) 설비 가용, (4) 납기 여유.** 모두 통과 시 WO 발행 → 자재 자동 예약 → POP 화면 노출(INJ/IMG/PNT).



4-Check Gate

순차 검증, 1개라도 실패 시 차단.
P0·MRP·설비·납기



Auto Reserve

발행 시 자재 자동 예약.
BR-PP-003



Equip Warning

설비 Blocked 시 경고 + 보류 권고.
BR-PP-007



POP Exposure

발행 즉시 현장 POP 표시.
INJ/IMG/PNT

관련 BR ▶

BR-PP-001

BR-PP-002

BR-PP-003

BR-PP-007

PP-CAL



Production Calendar

Web PC

2nd Dev

L3 DETAIL →

생산 캘린더 — 월/주
시각화



What you'll see: 전체 WO를 월/주 캘린더로 시각화 — 색상 = 라우팅·고객사, 빨강 = D-3 임박
드래그로 일정 재배정 (라인장 PIN 필요)

INPUT

All WOs
(PP-04)

OUTPUT

Visual
Schedule +
Drag

UI

Month /
Week
Toggle

HIGHLIGHT

D-3 빨강

EN · DESCRIPTION

Visual monthly/weekly calendar of all Work Orders with due dates. D-3 items are highlighted in red. Drag-to-reschedule supported (supervisor PIN required). Filters by routing (A/B), customer (SAV/GEO), and status. Click event to drill into PP-04 WO detail.

KO · 설명

전체 WO를 월/주 단위 캘린더로 시각화. D-3 이내 항목 빨간색 강조. 드래그 일정 조정 지원(라인장 PIN). 라우팅·고객사·상태 필터. 이벤트 클릭 시 PP-04 WO 상세 진입.



Month / Week View

월/주 단위 토글.
FullCalendar.js



Color Code

라우팅 A/B + 고객사 SAV/GEO 색상.
시각 분류



D-3 Red

납기 임박 빨간 강조 (BR-008).
놓침 방지



Drag Reschedule

드래그로 일정 변경 (PIN).
감사 기록

관련 BR >

BR-PP-008

PP-OTD



On-Time
Delivery
KPI

Web PC

2nd Dev

L3 DETAIL →

납기 준수를 현황 —
실시간 대시보드

What you'll see: OTD 대시보드 — 고객사·라우팅·기간별



정시/지연 비율 + 지연 원인 분석 (자재·설비·품질)

실시간 갱신, 95% 목표 대비 실적 추적

INPUT

FG
Shipment +
WO

OUTPUT

KPI
Dashboard
+ Cause

TARGET

≥ 95%

DRILL

지연 원인 자
동 분류

EN · DESCRIPTION

Real-time OTD KPI dashboard
by customer (SAV/GEO), routing
(A/B), and time period
(day/week/month). Shows on-
time vs delayed deliveries with
root-cause attribution (material
shortage, equipment downtime,

KO · 설명

고객사(SAV/GEO)·라우팅(A/B)·기
간(일/주/월)별 OTD 실시간 대시보
드. 정시/지연 비율, 지연 원인(자재
부족·설비 정지·품질 재작업 등) 분
석 제공. 목표 ≥ 95%. 지연 WO 목
록 드릴다운.

quality rework, etc.). Target $\geq$ 95%. Drill-down to delayed WO list.



Multi-Dimension

고객·라우팅·기간 3축 필터.
조합 분석



Root Cause

자연 원인 자동 분류·집계.
자재/설비/품질



Trend Chart

시계열 OTD 추이.
목표선 95%



Delay Drill

자연 WO 목록 드릴다운.
PP-04 진입

관련 BR ▶ 참조: BR-PP-008 (D-3 알림)

PP-LSB



Line Schedule Board

라인별 분단위 스케줄
보드

Web PC

2nd Dev

L3 DETAIL →



What you'll see: WO를 각 라인의 고유 시간패턴(MD-29)
위에 배치하는 분단위 간트 — 가동/계획비가동 밴드 + 캐파 부
하율

가동량 초과·계획비가동 침범 시 배치 차단

INPUT

MD-29
Pattern +
WO

OUTPUT

PP_LineSchedule
+ POP

GRAIN

1-minute
slot

AUTO

Auto-fill
by D-Day

EN · DESCRIPTION

KO · 설명

Minute-level Gantt board scheduling Work Orders onto each line's own time pattern. Lines render their MD-29 operating / planned-down bands; WO blocks drop only onto operating windows. Capacity = Σ operating minutes; overflow blocks Publish.

작업지시를 라인별 고유 시간패턴 위에 배치하는 분단위 간트 보드. MD-29 가동·계획비가동 밴드 렌더링, WO 블록은 가동창에만 배치. 캐파 = Σ 가동분, 초과 시 발행 차단.



Per-line Pattern

라인별 MD-29 시간패턴 기준.
공용 캘린더 아님



Window Fit

WO는 단일 가동창 내 배치.
BR-PP-031



Capacity Load

Σ 가동분 대비 부하율.
100% 초과 차단



Publish

POP 푸시 + ODM 기준선.
BR-PP-033

관련 BR ▶

BR-PP-030

BR-PP-031

BR-PP-032

PP-ODM

Operating / Downtime



Monitor

가동·계획비
가동·실제비
가동 실시간
모니터

Web PC / 현황판

2nd Dev

L3 DETAIL →

What you'll see: 라인별 계획 패턴(MD-29) vs 실제 설비 신호를 실시간 대조 — 매 분(分)을 가동/계획비가동/실제비가동 3상태로 분류

계획상 가동인데 정지 시 UNPLANNED 자동 판정 + 사유 입력 요청

INPUT

MD-29 +
설비 신호

OUTPUT

PP_LineStateLog

REFRESH

WebSocket
1-min

ALERT

비가동 10
분 초과

EN · DESCRIPTION

Real-time wallboard comparing each line's planned MD-29 pattern against actual equipment signal. Each minute resolves to OPERATING / PLANNED_DOWN / UNPLANNED_DOWN; a stop inside an operating window auto-classifies as unplanned and prompts a PP-DTL reason.

KO · 설명

라인별 계획 MD-29 패턴과 실제 설비 신호를 실시간 대조하는 현황판. 매 분을 가동/계획비가동/실제비가동으로 판정, 가동창 내 정지는 실제비가동 자동분류 + PP-DTL 사유 입력 요청.



3-State Classify

가동·계획비가동·실제비가동.
분단위 판정



Unplanned Detect

가동창 내 정지 자동분류.
BR-PP-035



Reason Prompt

미입력 비가동 PP-DTL 연계.
사유 강제



Operating Rate

가동률 실시간 집계.
PP-OEE 피드

관련 BR ▶

BR-PP-035

BR-PP-036

BR-PP-037

PP-OEE



Line
Availability
& OEE

Web PC

2nd Dev

L3 DETAIL →

What you'll see: PP_LineStateLog를 시간손실 워터폴로 롤업 — 부하시간 → 가동 → OEE = 가용성×성능×품질
계획/실제 비가동을 분리해 라인 고유 패턴을 공정 반영

INPUT

PP_LineStateLog

OUTPUT

PP_LineOEE
snapshot

PERIOD

Day /
Week /
Month

TARGET

OEE ≥
75%

EN · DESCRIPTION

Rolls up the classified state log into a time-loss model and computes per-line OEE. Loading Time minus planned + unplanned downtime gives Availability; combined with Performance and Quality it yields $OEE = A \times P \times Q$, snapshotted per period.

KO · 설명

분류된 상태 로그를 시간손실 모델로 롤업하여 라인별 OEE 산출. 부하시간 - 계획·실제 비가동 → 가용성, 성능·품질과 결합해 $OEE = A \times P \times Q$, 기간별 스냅샷 저장.



Loss Waterfall

부하→가동→가치가동 손실 분해.
시각 분석



Planned Excluded

가용성은 실제 비가동만 손실.
BR-PP-041



Period Snapshot

기간별 OEE 고정 스냅샷.
감사 추적



RPT Feed

RPT-06/07로 전달.
OEE-KPI

관련 BR ▶

BR-PP-042

BR-PP-041

BR-PP-044

PP-DTL

Downtime



Log Entry

비가동 로그 입력
— POP 터치패널

POP / PDA

2nd Dev

L3 DETAIL →



What you'll see: PP-ODM이 감지한 실제 비가동 이벤트에
작업자가 사유(MD-25)·4M1E 원인(MD-22)을 터치 입력
사유 기록 후 이벤트 종료 → PP-OEE 가용성 손실 귀속

INPUT

ODM
Downtime
Event

OUTPUT

PP_LineDowntimeLog

REASON

MD-25 ·
MD-22

DEVICE

POP
15" 터
치

EN · DESCRIPTION

POP touch-panel screen where the operator records the reason for an unplanned downtime event detected by PP-ODM. Taps a downtime reason (MD-25) and optional 4M1E cause (MD-22); the closed event flows to PP-ODM and PP-OEE.

KO · 설명

PP-ODM이 감지한 실제 비가동 이벤트의 사유를 작업자가 기록하는 POP 터치패널. 비가동 사유(MD-25)·4M1E 원인(MD-22) 터치 선택, 종료된 이벤트는 PP-ODM·PP-OEE로 전달.



Reason Tiles

MD-25 DOWNTIME 사유 터치.
대형 타일



4M1E Cause

MD-22 원인 분류 선택.
선택 입력



Comment Gate

필수 사유는 코멘트 강제.
BR-PP-046



MNT Sync

고장 사유는 MNT-06 동기화.

관련 BR ▶

BR-PP-045

BR-PP-046

BR-PP-048

▼ BUSINESS RULES · 업무 규칙 (PP · 16건)

Rule ID	Rule (EN)	규칙 (KO)	조건 · Condition	Action
BR-PP-F001	Warn on unmapped customer item no.	고객사 품번 미 매핑 시 경고	CustomerItemNo 매핑 없음	Warn
BR-PP-F002	Alert on Forecast vs Order gap > 20%	Forecast 대비 수 주 편차 20% 초과 알림	Gap > Forecast × 0.2	Alert
BR-PP-F003	Retain history on Forecast re-upload	Forecast 재업로드 시 이력 보관 후 UPDATE	동일 월·품목 중복	Auto
BR-PP-001	Block WO if Phase-0 incomplete	Phase-0 미완료 WO 생성 차단	Phase0Complete = 0	Block
BR-PP-002	Block WO on MRP shortage	MRP 재고 부족 WO 자동 차단	ShortageQty > 0	Block
BR-PP-003	Auto-reserve material on WO release	WO 발행 시 자재 자동 예약	Status = Released	Auto
BR-PP-004	Release reserved qty on WO cancel	WO 취소 시 예약 수량 자동 해제	Status = Cancelled	Auto
BR-PP-005	MRP auto-run nightly at 00:00	MRP 야간 자동 실행	매일 00:00 배치	Auto
BR-PP-006	Retry PR SAP B1 transmission ×3	PR SAP B1 전송 실패 시 3회 재시도	전송 오류	Retry
BR-PP-007	Warn if equipment blocked on WO release	WO 발행 전 설비 가용 상태 확인	Equipment.Status = Blocked	Warn

Rule ID	Rule (EN)	규칙 (KO)	조건 · Condition	Action
BR-PP-008	Alert DDay ≤ 3 WO	납기 D-3 이내 WO 강조 알 림	DueDate - TODAY ≤ 3	Alert
BR-PP-030	Each line schedules on its own MD-29 time pattern	각 라인 은 고유 MD-29 시간패턴 으로 스 케줄링	PP-LSB 일정 수립	Auto
BR-PP-031	WO block must fit one OPERATING window	WO 블록 은 단일 가동창 내 완전 배치	가동창 초과·침범	Block
BR-PP-035	Stop in operating window → UNPLANNED_DOWN	가동창 내 설비 정지는 실제비가 동 자동 분류	설비 신호 STOP	Auto
BR-PP-042	OEE = Availability × Performance × Quality	OEE = 가용성 × 성능 × 품질, 기간별 스냅샷	PP-OEE 롤업	Auto
BR-PP-045	Unplanned downtime must close with a reason code	실제 비 가동은 사유코드 로 종료 필수	비가동 이벤트 종료	Block

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VOL.4 · PP MODULE

PRODUCTION PLAN

생산계획

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PP-03Plan & Confirm생산계획 수립

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VOL.4 · PP · PRODUCTION PLANNING · FORECAST 관리

PP-01FORECAST MANAGEMENT

FORECAST

DEVICEWeb PC

PHASE1st Dev

ROLEPLAN Manager

FREQMonthly · 월간

URL/pp/forecast

01Overview

화면 개요 · 목적 · 컨텍스트

Manages customer forecast intake (SAV/GEO). Excel files downloaded from customer SCM portals are uploaded, validated, and mapped against the PP_CustomerItemMap table to convert customer item numbers to A-MES item codes. Provides M+1 to M+6 monthly demand visibility with M+1~M+3 binding for production planning and M+4~M+6 as reference for capacity planning. Detects gaps between forecast and confirmed orders, alerting PLAN/MGR when deviation exceeds 20%.

고객사(SAV/GEO) Forecast 수신·관리 화면. 고객 SCM 포털에서 다운로드한 Excel을 업로드하면 PP_CustomerItemMap으로 고객사 품번 → A-MES 품목코드 매핑·검증. M+1~M+3은 생산계획 구속력, M+4~M+6은 능력계획 참고. Forecast 대비 확정 수주 편차 20% 초과 시 PLAN/MGR 알림.

SCREEN ID	PP-01
LAYOUT	Upload + Forecast table + Gap analysis 업로드·표·편차분석
SOURCE	SCM portal Excel (manual) SCM 포털 Excel
FUTURE	SCM Portal API (Phase 2)

TABLES	PP_Forecast , PP_ForecastHistory , PP_CustomerItemMap
RBAC	PP.Forecast

02 Screen Mockup

화면 목업 — 주요 UI 구성

PP-01 · FORECAST MANAGEMENT · WEB PC

A-MES · PP · Forecast Management

PP > Forecast

All Customers

2025-Q4

Item Code...

Search

Upload Excel

Export

142

Mapped Items

매핑 품목

3

Unmapped

미매핑

8

Gap >20%

편차 20% ↑

2025-08-01

Last Upload

최근 업로드

Customer 고객사	Customer Item No. 고객사 품번	A-MES Code 내부 품목	Item Name 품목명	M+1	M+2
SAV	SAV-TRIM-LH	TRIM-LH-001	도어트림 LH	2,400	2,200
GEO	GEO-TRIM-RH	TRIM-RH-001	도어트림 RH	1,600	1,500
SAV	SAV-CONS-UP	CONS-UP-001	콘솔어퍼	800	900
GEO	GEO-NEW-PT	— △	UNMAPPED · 매핑 필요	500	500

03 UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-01	Customer Filter 고객사 필터	SELECT	N	Options: All / SAV / GEO. Affects all downstream listings.	All
CMP-02	Period Selector 기간 선택	SELECT	N	Quarter/Month selector. Loads forecast for selected period.	Current Q
CMP-03	Upload Excel Excel 업로드	FILE	Y	Accepts .xlsx with required columns: Customer Item No, M+1, M+2, M+3, M+4~6.	—
CMP-04	Validate Mapping 매핑 검증	AUTO	—	Auto-runs on upload. Joins with <code>PP_CustomerItemMap</code> . Unmapped rows flagged.	—
CMP-05	Gap Calculation 편차 계산	AUTO	—	$\text{Gap\%} = (\text{Confirmed} - \text{Forecast}) / \text{Forecast} \times 100$. >20% → BR-PP-F002 alert.	—
CMP-06	Export Excel Excel 출력	BUTTON	N	Exports current view with gap analysis.	—

Event & Action

04 Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER click · [📎 Upload Excel]

User selects Excel file from customer SCM portal.

고객 SCM 포털에서 다운로드한 Excel 선택.

VALIDATE Schema check: required columns present, no empty Item No.

2

SYSTEM parse · file → rows

System parses rows and matches against `PP_CustomerItemMap` for item code mapping.

행을 파싱하여 `PP_CustomerItemMap`으로 품번 매핑.

EFFECT Mapped: insert/update `PP_Forecast`.

3

Unmapped: flag in warning list.

SYSTEM calc · gap analysis

For each mapped item, fetch confirmed orders from PP_SupplyPlan and compute gap%.

매핑된 품목별로 PP_SupplyPlan의 확정 수주와 비교하여 편차% 계산.

ALERT Gap > 20% → BR-PP-F002 alert sent to PLAN/MGR via email.

4

SYSTEM history · re-upload

If duplicate month+item rows, move existing to PP_ForecastHistory before update.

동일 월·품목 중복 시 기존 행을 PP_ForecastHistory로 이관 후 갱신.

AUDIT History retained 3 years per audit policy.

5

USER click · row [Map] (for unmapped)

User clicks Map button on unmapped row, opens dialog to manually link customer item no to A-MES item code.

미매핑 행의 [Map] 클릭 시 수동 매핑 다이얼로그 표시.

API `POST /api/pp/customer-item-map` · adds mapping for future uploads.

6

SYSTEM complete · forecast saved

Forecast tab refreshed. Downstream PP-03 plan creation can now reference these qty.

Forecast 탭 갱신. PP-03 생산계획 수립에서 이 수량 참조 가능.

DOWNSTREAM PP-03 (planning), PP-05 (MRP capacity check)

05

Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP01-001

Unmapped Item

미매핑 품번

CAUSE Customer item no not found in `PP_CustomerItemMap`.

DETECT On upload parse.

ACTION	Row flagged red, excluded from save. Warning list shown.
RECOVER	User maps via [Map] button or MD team updates mapping master.

⚠ UNMAPPED · MAP REQUIRED

E-PP01-002

Invalid File Format

파일 형식 오류

CAUSE	Excel missing required columns or wrong type.
DETECT	Schema validation on upload.
ACTION	Reject entire upload. Show specific missing columns.
RECOVER	Use [Sample Template] download.

❌ INVALID FORMAT

E-PP01-003

Forecast Gap Alert

편차 알림

CAUSE	Confirmed orders vs forecast deviation > 20%.
DETECT	Auto on save.
ACTION	Email alert to PLAN+MGR. Item flagged in table.
RECOVER	Review with sales/customer.

i GAP ALERT — REVIEW

E-PP01-004

Concurrent Upload

동시 업로드 충돌

CAUSE	Two users upload same month forecast simultaneously.
DETECT	Row lock on PP_Forecast.
ACTION	Second user blocked with current state shown.
RECOVER	Reload and re-apply changes if still needed.

⚠ CONCURRENT — RELOAD

E-PP01-005

History Rollback Required

이력 복구 필요

CAUSE	User requests rollback to previous forecast version.
DETECT	Manual via [History] tab.

ACTION	Requires manager approval. Restored from <code>PP_ForecastHistory</code> .
RECOVER	Manager approves; new entry inserted (not in-place update).

⚠ ROLLBACK · APPROVAL

E-PP01-006

Empty Upload

빈 파일

CAUSE	Excel has zero data rows.
DETECT	Pre-parse check.
ACTION	Block save. Suggest using correct template.
RECOVER	Re-download from customer portal.

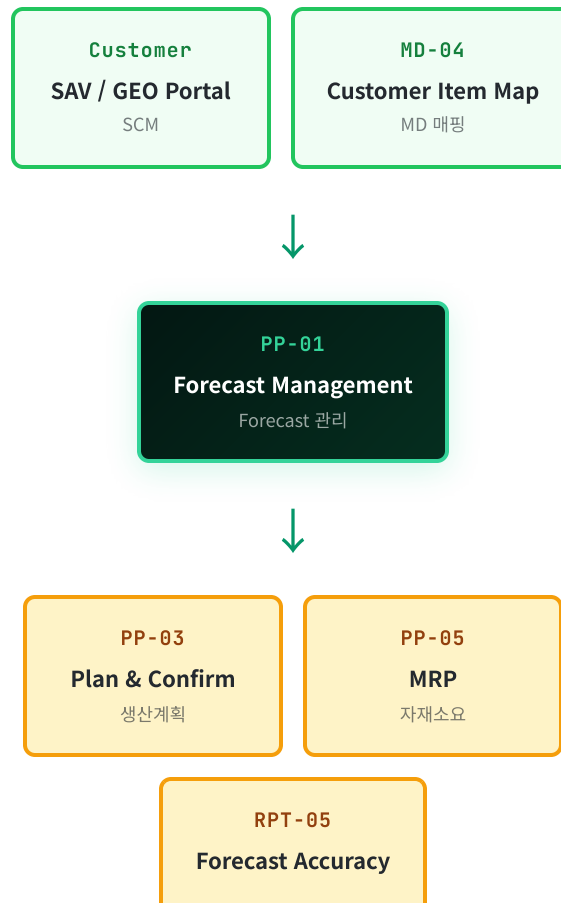
⚠ NO DATA

06

Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



07 Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-F001	Warn on unmapped customer item no. on Forecast upload.	Forecast 업로드 시 미매핑 고객사 품번 경고.	ALT
BR-PP-F002	Alert when Forecast vs confirmed order gap exceeds 20%.	Forecast 대비 확정 수주 편차 20% 초과 시 알림.	ALT
BR-PP-F003	On Forecast re-upload, retain history before overwriting.	Forecast 재업로드 시 이력 보관 후 UPDATE.	AUT

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VOL.4 · PP MODULE

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PP-06Purchase Req.자재발주 요청

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VOL.4 · PP · PRODUCTION PLANNING · SAP B1공급 계획 수신

PP-02

SAP B1 SUPPLY PLAN IMPORT

S A P I M P O R T

DEVICEWeb PC

PHASE1st Dev

ROLEPLAN Manager

FREQDaily · 일간

URL/pp/supply-plan/import

01Overview

화면 개요 · 목적 · 컨텍스트

Receives confirmed supply plans from SAP Business One.
Current (Phase 1): manual Excel import. Future (Phase 2): direct integration via SAP B1 Service Layer API. Each supply plan row corresponds to a confirmed sales order with order qty, due date, and customer reference. Once imported, plans flow to PP-03 for feasibility review and WO batch creation.

SAP B1에서 확정된 공급계획 수신. 1차: Excel 수동 import. 2차: SAP B1 Service Layer API 직접 연동. 각 행은 확정 판매오더(고객사·수량·납기). Import 후 PP-03에서 가용성 검토·WO 일괄 생성으로 연계.

SCREEN ID	PP-02
SOURCE	SAP B1 Excel export (Phase 1)
FUTURE	Service Layer APIGET /Orders
SCHEDULE	Daily 07:00 KST batch매일 07:00 배치
TABLES	PP_SupplyPlan, PP_SupplyPlanDetail
RBAC	PP.Plan

02

Screen Mockup

화면 목업 — 주요 UI 구성

● PP-02 · SAP B1 SUPPLY PLAN IMPORT · WEB PC

A-MES · PP · SAP B1 Supply Plan Import PP > Supply Plan

Import Method: Excel Upload (Phase 1) ▼

파일 선택 선택된 파일 없음 ↑ Import

📄 Sample Template

48 Today's SOs 금일 SO	46 Confirmed 확정	2 Open 미확정	07:12 Last Sync 동기화
----------------------------	-----------------------	------------------	---------------------------

SO No.	Customer 거래선	Item Code	Item Name	Order Qty	Due Date	Status
SO-2025-08001	SAV	TRIM-LH-001	도어 트림 LH	2,350	2025-09-15	Confirmed
SO-2025-07055	SAV	TRIM-RH-001	도어 트림 RH	1,580	2025-08-31	Confirmed
SO-2025-08003	GEO	CONS-UP-001	콘솔 어퍼	800	2025-09-25	Open

03

UI Components

Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증·동작	DEFAULT
CMP-01	Import Method 수신 방식	SELECT	Y	Excel Upload (Phase 1) / API (Phase 2, disabled until ready).	Excel
CMP-02	File Picker 파일 선택	FILE	Y	.xlsx file from SAP B1 export. Required: SO No,	—

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
				Customer, Item, Qty, Due.	
CMP-03	[↑ Import] 가져오기	BUTTON	Y	Triggers parse + insert into PP_SupplyPlan + Detail . Audit log entry.	—
CMP-04	Sample Template 샘플 양식	BUTTON	N	Downloads blank Excel with required column headers.	—
CMP-05	Daily Auto-Sync 일 자동 동기화	SCHEDULED	—	07:00 KST batch. Phase 2 only (API). Bypasses Excel.	Daily 07:00

04 Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

SCHEDULED cron · 07:00 daily sync

[Phase 2 future] System auto-calls SAP B1 Service Layer to fetch /Orders?filter=DocStatus eq 'O' and inserts new SOs.

[2차 예정] 매일 07:00 SAP B1 Service Layer /Orders 호출하여 미확정 SO 자동 import.

API GET /b1s/v1/Orders?
\$filter=DocStatus%20eq%20%27O%27

2

USER click · [↑ Import]

Manager uploads SAP B1 Excel export. System validates schema and parses rows.

관리자가 SAP B1 Excel export 업로드. 스키마 검증 + 행 파싱.

VALIDATE Required cols: SO No, Customer, Item Code, Qty, Due Date, Status

3

SYSTEM save · transactional insert

Inserts header row in PP_SupplyPlan and detail rows in PP_SupplyPlanDetail atomically.

PP_SupplyPlan 헤더 + PP_SupplyPlanDetail 상세를 원자적으로 INSERT.

```
DB BEGIN TRAN; INSERT header; INSERT details;
COMMIT
```

4

SYSTEM match · forecast vs SO

Cross-references confirmed SO against PP-01 Forecast for the same item/period.

확정 SO와 PP-01 Forecast를 품목·기간별로 매칭.

ALERT Gap > 20% triggers BR-PP-F002 alert (same as PP-01).

5

SYSTEM complete · ready for PP-03

Imported SOs appear in PP-03 with Forecast reference and FG stock context.

Import된 SO가 PP-03에서 Forecast 참조·FG 재고와 함께 표시.

DOWNSTREAM PP-03 plan creation, calendar overlay

05 Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP02-001

Schema Mismatch

스키마 불일치

CAUSE	Excel column names don't match expected.
DETECT	Pre-parse header check.
ACTION	Block import. Show missing/extra columns.
RECOVER	Re-download SAP B1 export or use Sample Template.

 COLUMN MISMATCH

E-PP02-002

Duplicate SO No

SO 중복

CAUSE	SO No already exists in PP_SupplyPlan.
DETECT	Unique key constraint.
ACTION	Show duplicate count, ask user: update or skip.
RECOVER	Update overwrites Qty/Due Date.

⚠️ DUPLICATE SO

E-PP02-003

SAP API Timeout

SAP API 타임아웃

CAUSE	[Phase 2] Service Layer call exceeds 30s.
DETECT	Heartbeat monitor.
ACTION	Retry 3× with exponential backoff. Then alert.
RECOVER	Manual Excel fallback or wait for SAP recovery.

🚫 SAP TIMEOUT

E-PP02-004

Unknown Item Code

미등록 품목

CAUSE	SO item not in <code>MD_Item</code> master.
DETECT	Per-row validation.
ACTION	Skip row, log to error report. Other rows continue.
RECOVER	MD team adds item or removes invalid SO in SAP.

🚫 ITEM NOT IN MD

E-PP02-005

Past Due Date

과거 납기

CAUSE	SO due date is in the past.
DETECT	Date validation.
ACTION	Allow import but flag yellow. Cannot create WO from past-due SO without override.
RECOVER	Update due date in SAP B1.

⚠️ PAST DUE

E-PP02-006

Authentication Failed

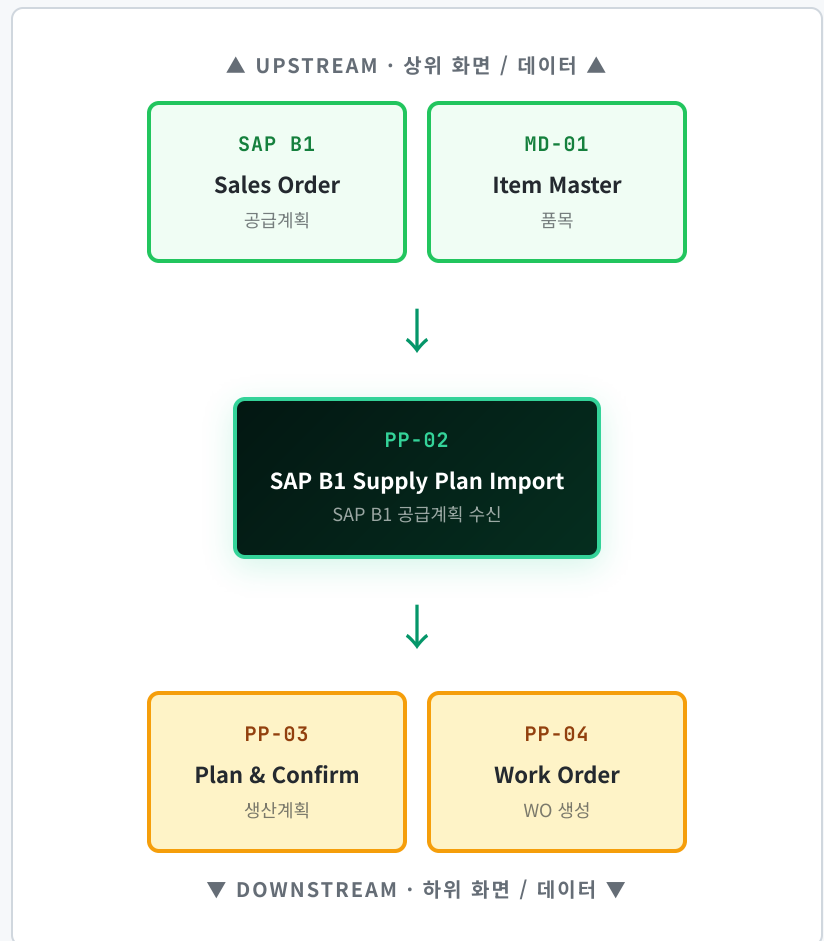
인증 실패

CAUSE	[Phase 2] SAP B1 Service Layer login token expired.
DETECT	API 401 response.
ACTION	Re-authenticate via stored credentials. Retry once.
RECOVER	SYS admin updates SAP credential secret.

06

Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름



07

Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-F002	Alert when Forecast vs confirmed order gap exceeds 20%.	Forecast 대비 확정 수주 편차 20% 초과 시 알림.	ALT

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-001	Block WO creation if Phase-0 master data is incomplete.	Phase-0 기준정보 미완료 시 WO 생성 차단.	BLK

A-MES · VOL.4 · PP-02 SAP B1 Supply
Plan Import · L3 Detail Spec v1.0 ·
Bilingual EN/KO

Automotive Parts MES ·
SAV & GEO Supply ·
Confidential

A-MESIndex

WH WarehousePP PlanINJ InjectionIMG WrappingPNT PaintingQC QualityFG Finished GoodsMNT Main

L1A-MES Index > L2VOL.4 · PP Production Plan > L3PP-03 · Plan & Confirm

VOL.4 · PP MODULE

PRODUCTION PLAN

생산계획

L3 SCREENS · 화면 상세

PP-01Forecast MgmtForecast 관리

PP-02SAP B1 Import공급계획 수신

PP-03Plan & Confirm생산계획 수립

PP-04Work Order Mgmt작업지시 관리

PP-05MRP자재 소요 계획

PP-06Purchase Req.자재발주 요청

PP-07WO ReleaseWO 발행

PP-CALProduction Calendar생산 캘린더

PP-0TDDelivery KPI납기 현황

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05Exception Handling

06Linked Screens

07Business Rules

VOL.4 · PP · PRODUCTION PLANNING · 생산 계획 수립 및 확인

PP-03

PLAN & CONFIRM

생 산 계 획

DEVICE Web PC

PHASE 1st Dev

ROLE PLAN Manager

FREQ Daily · 일간

URL /pp/pPlan

01Overview

화면 개요 · 목적 · 컨텍스트

PLAN manager reviews supply plans against FG inventory, equipment load, and forecast reference. For each plan line, the system calculates Net Required Qty (= Order Qty – FG Stock), shows current line load %, and compares vs forecast. Items selected via checkbox are batch-released to PP-04 as Work Orders.

PLAN 담당자가 공급계획을 완제품 재고·설비 부하·Forecast와 대조하여 생산 가능성 검토. 행별로 순생산 필요량(Order – FG Stock) 자동 계산, 라인 부하율 표시, Forecast 대비. 체크박스로 선택된 행은 PP-04 WO 일괄 생성으로 전송.

SCREEN ID	PP-03
SOURCE	PP-02 imported supply plans
CALC	Net Req = Order – FG Stock
CAPACITY	MNT equipment load (real-time)
TABLES	PP_SupplyPlan , WH_Inventory (FG), MNT_Equipment
RBAC	PP.Plan

02

Screen Mockup

화면 목업 — 주요 UI 구성

● PP-03 · PLAN & CONFIRM · WEB PC

A-MES · PP · Plan & Confirm PP > Plan

2025-09-15 ~ 2025-09-30

All Lines

☐	SO No.	Item	Order 공급	FG Stock 완제품	Net Req. 순생산	Forecast	Line 설비
☒	SO-2025-08001	TRIM-LH-001 도어트림 LH	2,350	320	2,030	2,400	72%
☐	SO-2025-08003	CONS-UP-001 콘솔어퍼	800	150	650	900	45%
☐	SO-2025-08009	TRIM-RH-002 도어트림 RH (신규)	1,200	0	1,200 ⚠	—	Ph 미

03

UI Components
Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-01	Date Range 기간 선택	DATE×2	Y	From-To. Default: this week + next week. Max 90 days.	Week+1

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증·동작	DEFAULT
CMP-02	Line Filter 라인 필터	SELECT	N	Filter by production line: INJ-L1/L2, IMG-L1, PNT-L1.	All
CMP-03	Plan Checkbox 계획 체크박스	CHECKBOX	Y	Per-row selection for batch WO creation. Disabled if blocking conditions.	Unchecked
CMP-04	Net Req Calc 순생산 계산	AUTO	—	Net Req = Order Qty - FG Stock . Real-time from WH_Inventory.	—
CMP-05	Line Load Display 설비 부하 표시	AUTO	—	Bar showing current line utilization %. Red >90%, yellow 70-90%, green <70%.	—
CMP-06	[▶ Batch Create WO] WO 일괄 생성	BUTTON	Y	Creates Draft WOs for all checked rows. Validates Phase-0 first.	—
CMP-07	Calendar View 캘린더 보기	BUTTON	N	Switches to PP-CAL calendar view, preserving filter.	—

04 Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER page.load · open PP-03

Loads all confirmed SOs from PP_SupplyPlan with status Open in date range. Calculates Net Req for

each.

기간 내 확정 SO 로드 + 행별 Net Req 자동 계산.

API `GET /api/pp/plan?from=...&to=...`

2

SYSTEM `fetch · FG inventory`

Real-time fetch from WH_Inventory for each item's FG (finished goods) on-hand qty.

각 품목별 완제품 재고를 WH_Inventory에서 실시간 조회.

QUERY `SELECT item_code, SUM(on_hand) FROM
wh_inventory WHERE warehouse='WH-FG' GROUP BY
item_code`

3

SYSTEM `fetch · line load`

Queries MNT_Equipment for current capacity utilization per line.

MNT_Equipment에서 라인별 현재 가동률 조회.

EFFECT Each row shows line load bar

4

USER `check · row selection`

User selects rows by checkbox. System dynamically updates batch size and total qty in footer.

행 체크 시 일괄 처리 건수·총 수량 푸터에 실시간 갱신.

UI Footer counter updates · disable conflicting rows

5

USER `click · [Batch Create WO]`

Triggers WO creation for all checked rows. Each becomes a Draft WO with routing per BOP (MD-04).

체크된 행에 대해 WO 일괄 생성. 각각 Draft 상태, BOP(MD-04) 기준 라우팅 자동 적용.

API `POST /api/pp/wo/batch` · returns array of wo_ids

6

SYSTEM `validate · Phase-0`

Before WO creation, validates that all required master data is complete (BOM/BOP/Inspection Standard).

WO 생성 전 기준정보 완성도 검증(BOM/BOP/검사 기준).

BLOCK If any missing → BR-PP-001 block · skip row · report

7

SYSTEM `redirect · to PP-04`

Successful batch redirects to PP-04 with new WOs highlighted.

성공 시 PP-04로 이동하며 신규 WO 하이라이트.

05

Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP03-001

Phase-0 Incomplete

Phase-0 미완료

CAUSE	Selected item has missing master data (BOM/BOP/Insp).
DETECT	Validation on Batch Create.
ACTION	Block creation for that row. Show specific missing master.
RECOVER	MD team completes master in MD-01~06.

⛔ PHASE-0 INCOMPLETE

E-PP03-002

Insufficient Capacity

능력 부족

CAUSE	Line load already > 95% for the period.
DETECT	Capacity check on select.
ACTION	Yellow warning. Allow creation but suggest scheduling later.
RECOVER	Adjust due date or split WO.

⚠ CAPACITY TIGHT

E-PP03-003

Negative Net Req

음수 순생산

CAUSE	FG stock exceeds Order Qty (overstock).
DETECT	Calc check.
ACTION	Show negative in green (good). Disable WO creation (no need to produce).
RECOVER	Skip this row.

ⓘ SUFFICIENT STOCK

E-PP03-004

BOP Not Defined

BOP 미정의

CAUSE	MD-04 routing not set for this item.
DETECT	JOIN check.
ACTION	Block creation. Cannot determine production sequence.
RECOVER	MD-04 BOP entry required.

🚫 NO BOP · SEE MD-04

E-PP03-005

Concurrent Edit

동시 편집

CAUSE	Same SO selected by another planner.
DETECT	Optimistic lock.
ACTION	Block save with current state of conflicting SO shown.
RECOVER	Refresh and re-select.

⚠️ CONCURRENT

E-PP03-006

Past Due Selected

과거 납기 선택

CAUSE	Due date already passed.
DETECT	Date check.
ACTION	Allow with confirmation. WO becomes overdue immediately.
RECOVER	Update SO due date in SAP first.

⚠️ ALREADY OVERDUE

06

Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲





07 Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-001	Block WO creation if Phase-0 master data is incomplete.	Phase-0 기준정보 미완료 시 WO 생성 차단.	BLK
BR-PP-002	Auto-block WO when MRP detects material shortage.	MRP 자재 부족 시 WO 자동 차단.	BLK
BR-PP-008	Alert PLAN/MGR when WO due date is within D-3.	WO 납기 D-3 이내 강조 알림.	ALT

VOL.4 · PP · PRODUCTION PLANNING · 작업지시 관리

PP-04

WORK ORDER MANAGEMENT

W O M G M T

DEVICEWeb PC

PHASE1st Dev

ROLEPLAN Manager

FREQContinuous · 상시

URL /pp/wo

01

Overview

화면 개요 · 목적 · 컨텍스트

Manages Work Orders through their full lifecycle: Draft → Released → In Progress → Closed. Each WO has a routing badge (A: INJ→IMG→QC→FG · B: INJ→PNT→QC→FG), production progress bar, D-Day indicator, and SAP B1 supply plan reference. Block reasons (Phase-0 incomplete, MRP shortage, equipment unavailable) are surfaced before release.

작업지시(WO) 전체 생명주기 관리: Draft → Released → In Progress → Closed. 각 WO는 라우팅 배지(A: INJ→IMG→QC→FG · B: INJ→PNT→QC→FG), 진척률 바, D-Day, SAP B1 공급계획 참조 표시. 발행 전 차단 사유(Phase-0 미완료·MRP 부족·설비 미가용) 표시.

SCREEN ID	PP-04
LIFECYCLE	Draft → Released → In Progress → Closed
ROUTING	A (INJ→IMG→QC→FG) · B (INJ→PNT→QC→FG)
DOWNSTREAM	INJ/IMG/PNT POP screens
TABLES	PP_WO , PP_WOMaterial
RBAC	PP.WO

02

Screen Mockup

화면 목업 — 주요 UI 구성

● PP-04 · WORK ORDER MANAGEMENT · WEB PC

A-MES · PP · Work Order Management PP > WO

All Status ▾ All Routing ▾

WO No / Item... Search + Manual WO

12 Draft	28 Released	15 In Progress	3 Blocked	5 D-3 Alert
-------------	----------------	-------------------	--------------	----------------

WO No.	Item	Plan Qty	Routing	Progress
WO-20250801-001	TRIM-LH-001 도어트림 LH	2,030	A: INJ→IMG→QC→FG	
WO-20250801-002	TRIM-LH-001-GY 도어트림 LH 회색	2,030	A	
WO-20250801-005	CONS-UP-001 콘솔어퍼 (PNT)	650	B: INJ→PNT→QC→FG	—

03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-01	Status Filter 상태 필터	SELECT	N	All / Draft / Released / In	ALL

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
				Progress / Closed / Blocked.	
CMP-02	Routing Filter 라우팅 필터	SELECT	N	A: INJ→IMG→QC→FG · B: INJ→PNT→QC→FG.	ALL
CMP-03	Search Box 검색	TEXT	N	WO no, item code, or SAP SO no. 300ms debounce.	(empty)
CMP-04	WO Progress Bar 진척률 바	AUTO	-	% = (Completed routings / Total routings). Color: green >80%, blue 40-80%, gray <40%.	-
CMP-05	D-Day Indicator D-Day	AUTO	-	Days = DueDate - TODAY . Red ≤ 3, yellow ≤ 7, green > 7.	-
CMP-06	Manual WO Create 수동 WO	BUTTON	N	Opens form to create ad-hoc WO not from SAP supply plan. Requires manager role.	-
CMP-07	Row Click → Detail 행 클릭	NAV	-	Opens WO detail panel with material allocation, routing, history.	-

Event & Action

04 Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER page.load · WO list

Loads all WOs with current status, progress, D-Day. Default sort by D-Day ascending.

전체 WO 목록 로드 (상태·진척률·D-Day). 기본 D-Day 오름차순.

API GET /api/pp/wo?date=today&sort=dday

USER click · row

2

Opens detail side panel with WO header, BOM allocation, routing steps, and history log.

행 클릭 시 상세 패널 표시 (헤더·BOM 할당·라우팅·이력).

API GET /api/pp/wo/:id/detail

3

SYSTEM push · WO status change

When downstream POP (INJ/IMG/PNT) updates routing step, WO progress and status auto-refresh via WebSocket.

POP 모듈(INJ/IMG/PNT) 공정 갱신 시 WebSocket으로 진척률·상태 자동 갱신.

PUSH WS event: wo.progress {wo_id, routing_step, qty}

4

SYSTEM alert · D-3 approaching

When WO due date is within 3 days, sends alert + highlights row red.

WO 납기 D-3 도달 시 알림 발송 + 행 빨간색 강조.

ALERT BR-PP-008 · email + in-app + row.class='rd'

5

USER click · [+ Manual WO]

Opens manual WO creation form. Requires manager role and provides bypass for non-SAP planning.

수동 WO 생성 폼. 매니저 권한 + SAP 미경유 케이스 처리.

API POST /api/pp/wo · source='manual', requires audit reason

6

USER click · row [Cancel]

User cancels Draft/Released WO. System auto-releases material reservations and reverses status.

Draft/Released WO 취소 시 자재 예약 자동 해제 + 상태 역전.

EFFECT BR-PP-004 · WH_Inventory.Reserved 감소

7

SYSTEM complete · status sync

WO status changes propagate to PP-CAL calendar and PP-OTD KPI calculations.

WO 상태 변경이 PP-CAL 캘린더 및 PP-OTD KPI 집계에 전파.

DOWNSTREAM PP-CAL refresh · PP-OTD recalc · RPT-05 dashboard

05

Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP04-001

MRP Block

MRP 차단

CAUSE	WO blocked by PP-05 MRP due to material shortage.
DETECT	Auto on MRP run.
ACTION	Row red. Status = Blocked. [Release] disabled. Show shortage detail.
RECOVER	Run PP-06 PR or wait for material.

🚫 BLOCKED · MRP

E-PP04-002

Cancel In Progress

진행중 취소

CAUSE	User attempts to cancel In Progress WO.
DETECT	Status check on cancel.
ACTION	Modal: 'Already in production. Cancel anyway?' Manager approval required.
RECOVER	Manager confirms; partial production logged.

⚠️ IN PROGRESS · APPROVE

E-PP04-003

Routing Mismatch

라우팅 불일치

CAUSE	POP screen reports routing step inconsistent with BOP.
DETECT	Push event validation.
ACTION	Yellow warning. Allow but log as anomaly. QC may need review.
RECOVER	Audit log entry + QC review.

⚠️ ROUTING DRIFT

E-PP04-004

Closed WO Edit Attempt

마감 WO 수정 시도

CAUSE	User tries to modify a Closed WO.
DETECT	Status check on save.
ACTION	Block. WO is immutable after Close.

RECOVER Create compensating WO if correction needed.

— CLOSED · IMMUTABLE

E-PP04-005

Manual WO Permission

수동 WO 권한

CAUSE Non-manager attempts Manual WO.

DETECT RBAC check.

ACTION Block with 'PP.WO.Manual role required'.

RECOVER Request role from sys admin.

— PERMISSION DENIED

E-PP04-006

WO Qty Mismatch

WO 수량 불일치

CAUSE Completed qty > planned qty.

DETECT Post-routing validation.

ACTION Yellow flag. Investigate overproduction reason.

RECOVER QC review + reconcile with SO.

⚠ OVER-PRODUCED

06 Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



PP-04

Work Order Management

작업지시 관리



▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

07 Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-002	Auto-block WO when MRP detects material shortage.	MRP 자재 부족 시 WO 자동 차단.	BLK
BR-PP-003	Auto-reserve materials on WO release.	WO 발행 시 자재 자동 예약.	AUT
BR-PP-004	Auto-release reserved qty on WO cancellation.	WO 취소 시 예약 수량 자동 해제.	AUT
BR-PP-008	Alert PLAN/MGR when WO due date is within D-3.	WO 납기 D-3 이내 강조 알림.	ALT

A-MESIndex

WH WarehousePP PlanINJ InjectionIMG WrappingPNT PaintingQC QualityFG Finished GoodsMNT Main

L1A-MES Index > L2VOL.4 · PP Production Plan > L3PP-05 · MRP – Material Requirements Planning

VOL.4 · PP MODULE

PRODUCTION PLAN

생산계획

L3 SCREENS · 화면 상세

PP-01Forecast MgmtForecast 관리

PP-02SAP B1 Import공급계획 수신

PP-03Plan & Confirm생산계획 수립

PP-04Work Order Mgmt작업지시 관리

PP-05MRP자재 소요 계획

PP-06Purchase Req.자재발주 요청

PP-07WO ReleaseWO 발행

PP-CALProduction Calendar생산 캘린더

PP-0TDDelivery KPI납기 현황

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03UI Components

04Event & Action Flow

05Exception Handling

06Linked Screens

07Business Rules

VOL.4 · PP · PRODUCTION PLANNING · 자재 소요 계획

PP-05

MRP – MATERIAL REQUIREMENTS PLANNING

M R P

DEVICE Web PC

PHASE 1st Dev

ROLE PLAN Manager · System

FREQ Daily 00:00 + on-demand 일 1회 + 수시

URL /pp/mrp

01Overview

화면 개요 · 목적 · 컨텍스트

Calculates material requirements by exploding BOM for each open WO and comparing against current stock and on-order quantities. Shortage = Required – Stock – OnOrder. Shortages over zero block the affected WOs (IsBlocked=1) and surface for PR generation in PP-06. Auto-runs nightly at 00:00; can be manually triggered for partial recalculation.

각 활성 WO의 BOM 분해 → 현재고·발주중 수량과 대조하여 자재 소요량 계산.
부족량 = 소요 – 재고 – 발주중. 부족 발생 시 해당 WO 차단
(IsBlocked=1)·PP-06 PR 생성 대상 목록 표시. 매일 00:00 자동 실행, 수동 부분 재실행 지원.

SCREEN ID	PP-05
SCHEDULE	Daily 00:00 KST + manual
FORMULA	Shortage = Req – Stock – OnOrder
EFFECT	WO.IsBlocked=1 on shortage > 0
TABLES	PP_WO , MD_BOM , WH_Inventory , WH_PurchaseOrder , PP_MRPLog

02 Screen Mockup

화면 목업 — 주요 UI 구성

● PP-05 · MRP — MATERIAL REQUIREMENTS PLANNING · WEB PC

A-MES · PP · Material Requirements Planning

PP > MRP

Last run: 2025-08-15 00:00:32 [▶ Run Now](#)

All WOs ▼

Item...

Export

87
Materials
Calc'd
계산 자재

71
Sufficient
충족

8
Shortage
부족

3
WO Blocked
차단 WO

Material	Required 소요	Stock 재고	OnOrder 발주중	Shortage 부족	Lead Time	Or Du 발 한
RM-PP-NAT-20 PP 펠렛 나뉠	3,600 kg	2,100 kg	500 kg	-1,000 kg	7 days	20 09
FAB-BK-001 원단(흑)	1,200 roll	800 roll	0	-400 roll	14 days	20 09
ADH-001 접착제	650 kg	980 kg	0	+330 kg	5 days	—
MAT-INJ-002 수지 B	2,000 kg	2,000 kg	0	±0	10 days	—

03 UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-01	Last Run Display 최근 실행	DISPLAY	—	Shows last successful MRP run timestamp with green indicator.	—
CMP-02	[▶ Run Now] 수동 실행	BUTTON	N	Triggers immediate MRP run. Locks UI until complete. Manager role required.	—
CMP-03	Scope Filter 범위 필터	SELECT	N	All WOs / Released only / Blocked only. Affects calculation scope.	All
CMP-04	Material Search 자재 검색	TEXT	N	Search by item code/name.	(empty)
CMP-05	Shortage Badge 부족 배지	AUTO	—	Red for shortage, gray for balanced, green for surplus.	—
CMP-06	[+ PR 생성] PR 생성	BUTTON	N	Per-row button enabled if shortage. Pre-fills PP-06 form with item, qty, due.	—
CMP-07	Excel Export 엑셀 출력	BUTTON	N	Exports full MRP result table.	—

04 Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

SCHEDULED cron · 00:00 daily MRP

Auto-runs nightly at 00:00 KST. Iterates all open WOs and explodes BOM via recursive query.

매일 00:00 KST 자동 실행. 활성 WO 전체에 대해 BOM 재귀 분해.

EFFECT BR-PP-005 · runtime 5-15 min · result in
PP_MRPLog

2

USER click · [▶ Run Now]

Manual MRP trigger. Useful after PP-03 plan changes or PP-02 import.

수동 MRP. PP-03 계획 변경 또는 PP-02 import 후 사용.

API **POST** /api/pp/mrp/run · returns job_id, polling for status

3

SYSTEM calc · BOM explosion

For each WO, recursively expand BOM to leaf-level materials. Apply WO.Qty × BOM.ChildQty.

WO별 BOM을 leaf까지 재귀 분해. WO.Qty × BOM.ChildQty로 소요량 누적.

DB Recursive CTE on **MD_BOM**

4

SYSTEM calc · shortage

Stock = WH_Inventory.AvailableQty (OnHand - Reserved). OnOrder = WH_PurchaseOrder.pending.
Shortage = Required - Stock - OnOrder.

Stock = WH_Inventory.AvailableQty (OnHand - Reserved).
OnOrder = WH_PurchaseOrder 미입고. 부족 = 소요 - 재고 - 발주중.

OUTPUT If > 0 → WO.IsBlocked=1 + insert **PP_MRPLog**

5

SYSTEM block · affected WOs

WOs with material shortage are auto-blocked. PLAN team notified.

부족 자재 보유 WO 자동 차단. PLAN팀에 알림.

EFFECT BR-PP-002 · WO.Status unchanged but IsBlocked=1

6

USER click · [+ PR 생성]

From shortage row, clicks PR. Pre-fills PP-06 form with item, shortage qty, required date.

부족 행에서 [PR 생성] 클릭 시 PP-06 폼에 자재·부족량·필요일 사전 입력.

NAV `/pp/pr?item=...&qty=...&due=...`

7

SYSTEM complete · ready for PR

MRP cycle complete. Shortages visible. PR generation ready in PP-06.

MRP 사이클 완료. 부족 자재 표시. PP-06에서 PR 생성 가능.

DOWNSTREAM PP-06 (PR), PP-04 (WO unblock after PO arrival)

05

Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP05-001

BOM Not Found

BOM 미정의

CAUSE	Item in WO has no BOM in MD-02.
DETECT	BOM lookup.
ACTION	Skip item, log error. WO marked partial calc.
RECOVER	MD team defines BOM.

🚫 NO BOM · SEE MD-02

E-PP05-002

Circular BOM Reference

BOM 순환참조

CAUSE	BOM contains cycle (A→B→A).
DETECT	Cycle detection.
ACTION	Halt calculation, alert MD team.
RECOVER	MD team fixes BOM.

🚫 CYCLE DETECTED

E-PP05-003

Long Runtime

장시간 실행

CAUSE	MRP run exceeds 15 min.
DETECT	Runtime monitor.
ACTION	Continue but flag. Investigate WO count or BOM depth.

RECOVER Optimize: archive Closed WOs.

⚠ SLOW MRP

E-PP05-004

Database Lock

DB 잠금

CAUSE Concurrent inventory update during MRP.

DETECT Lock timeout 60s.

ACTION Abort run, retry once. Then alert.

RECOVER Check WH transaction load.

🚫 DB LOCK · RETRY

E-PP05-005

Stale Inventory

재고 데이터 노후화

CAUSE WH_Inventory.last_updated > 1 hour ago.

DETECT Pre-run staleness check.

ACTION Yellow warning. Continue with current data.

RECOVER Check WH module health.

⚠ STALE STOCK

E-PP05-006

Massive Shortage

대규모 부족

CAUSE Shortage count exceeds 50 materials in single run.

DETECT Post-run summary.

ACTION Alert plant director. Likely supply chain issue.

RECOVER Convene S&OP meeting.

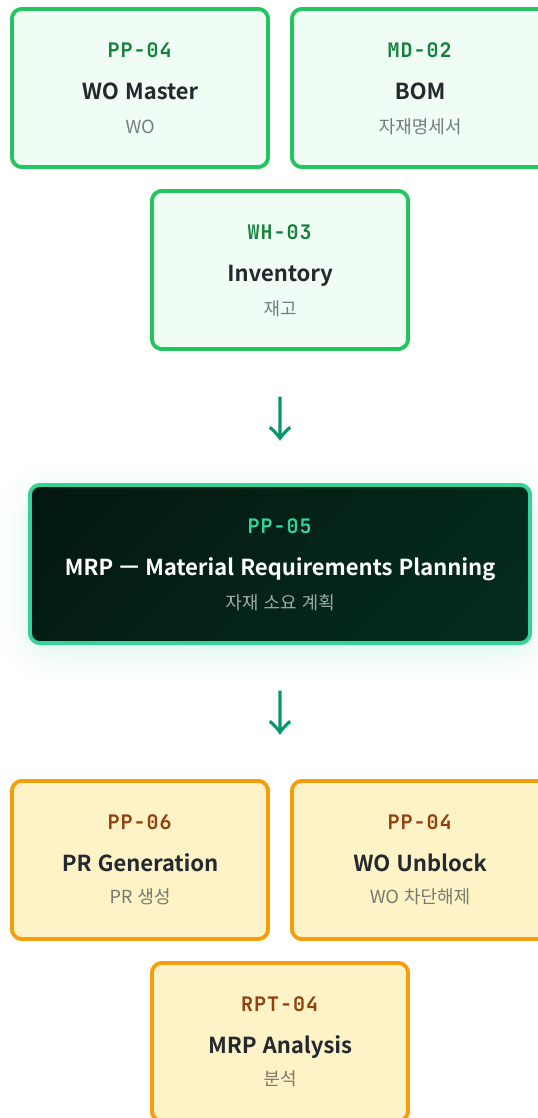
⚠ MASS SHORTAGE

06

Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

07 Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-002	Auto-block WO when MRP detects material shortage.	MRP 자재 부족 시 WO 자동 차단.	BLK
BR-PP-005	Auto-run MRP nightly at 00:00 KST.	매일 00:00 KST MRP 야간 자동 실행.	AUT

A-MES · VOL.4 · PP-05 MRP – Material
Requirements Planning · L3 Detail Spec
v1.0 · Bilingual EN/KO

Automotive Parts MES ·
SAV & GEO Supply ·
Confidential

A-MES

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PP Plan

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IMG Wrapping

PNT Painting

QC Quality

FG Finished Goods

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PP-06 · Purchase Requisition → SAP B1

VOL.4 · PP MODULE

PRODUCTION PLAN

생산계획

L3 SCREENS · 화면 상세

PP-01

Forecast Mgmt

Forecast 관리

PP-02

SAP B1 Import

공급계획 수립

PP-03

Plan & Confirm

생산계획 수립

PP-04

Work Order Mgmt

작업지시 관리

PP-05

MRP

자재 소요 계획

PP-06

Purchase Req.

자재발주 요청

PP-07

WO Release

WO 발행

PP-CAL

Production Calendar

생산 캘린더

PP-0TD

Delivery KPI

납기 현황

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Business Rules

VOL.4 · PP · PRODUCTION PLANNING · 자재 발주 요청서 (PR) 전송

PP-06

PURCHASE REQUISITION → SAP B1

P R S A P B 1

DEVICE Web PC

PHASE 1st Dev

ROLE PLAN Manager

FREQ As needed · 수시

URL /pp/pr

01

Overview

화면 개요 · 목적 · 컨텍스트

Generates Purchase Requisitions from MRP shortage results and transmits them to SAP B1 via Service Layer API (POST /PurchaseRequests). Tracks SAP B1 DocNum after successful transmission. Failures auto-retry 3× with backoff; persistent failure escalates to PLAN manager. Approved PRs flow to PO creation in SAP B1 and back to WH-01 (inbound schedule) as expected receipts.

MRP 부족 결과 기반 PR 생성 → SAP B1 Service Layer API(POST /PurchaseRequests) 전송. 성공 시 SAP B1 DocNum 추적. 실패 시 3회 자동 재 시도, 지속 실패 시 PLAN 관리자 에스컬레이션. 승인된 PR은 SAP B1에서 PO 생성 → WH-01 입고 예정으로 반영.

SCREEN ID	PP-06
ENDPOINT	POST /PurchaseRequests
RETRY	3× exponential backoff
DOWNSTREAM	SAP B1 PO → WH-01 inbound
TABLES	PP_PR , PP_PRSendLog

02 Screen Mockup

화면 목업 — 주요 UI 구성

● PP-06 · PURCHASE REQUISITION → SAP B1 · WEB PC

A-MES · PP · Purchase Requisition

PP > PR

All Status

연도 - 월 - 일

PR No / SAP DocNum...

Search

Send to SAP B1

12
Draft
작성중

8
Sent
전송완료

42
Approved · PO
승인·PO 생성

2
Failed
전송 실패

PR No.	SAP DocNum	Item	Qty	Required 필요일	Linked WO
PR-20250801-001	PR-B1-5542	RM-PP-NAT-20 원자재A	1,000 kg	2025-09-08	WO-20250801-001,002
PR-20250801-002	PR-B1-5543	FAB-BK-001 원단(흑)	400 roll	2025-09-01	WO-20250801-002
PR-20250801-003	—	RM-MB-GY-03 착색 MB	50 kg	2025-09-12	WO-20250801-005
PR-20250730-009	—	PNT-BK-NH578 블랙 도료	100 L	2025-08-25	WO-20250730-007

03 UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증·동작	DEFAULT
CMP-01	Status Filter 상태 필터	SELECT	N	Draft / Sent / Approved (PO) / Failed.	All
CMP-02	Date Filter 날짜 필터	DATE	N	Required date for PR.	(empty)
CMP-03	Search 검색	TEXT	N	PR No or SAP DocNum.	(empty)
CMP-04	[📧 Send to SAP B1] SAP 전송	BUTTON	Y	Sends selected Drafts via Service Layer. Updates status to Sent.	—
CMP-05	Retry Counter 재시도 카운터	AUTO	—	Display: 'Failed · n/3 retries'. Red after 3 fails.	—
CMP-06	[🔄 Retry] 재시도	BUTTON	N	Per-row retry for failed PRs. Resets retry counter.	—
CMP-07	[View PO] PO 보기	BUTTON	N	For approved PRs, deep-link to SAP B1 PO detail (opens in new tab).	—

04 Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER click · select Drafts

User multi-selects Draft PRs from list.

사용자가 Draft PR 다중 선택.

UI Checkbox column · footer shows selected count

2

USER click · [🚀 Send]

Triggers transmission. Each PR sent individually via API to SAP B1 Service Layer.

전송 트리거. 각 PR이 SAP B1 Service Layer API로 개별 전송.

API `POST /b1s/v1/PurchaseRequests` per row

3

SYSTEM transmit · retry on fail

On failure, auto-retries 3× with backoff (5s, 15s, 45s). After 3 failures, marks Failed.

실패 시 지수 backoff(5s/15s/45s)로 3회 자동 재시도. 3회 실패 후 Failed 상태.

EFFECT BR-PP-006 · log all attempts in `PP_PRSendLog`

4

SYSTEM track · DocNum

On success, SAP B1 returns DocNum. Stored in PP_PR.SAP_DocNum. Status = Sent.

성공 시 SAP B1이 DocNum 반환. PP_PR.SAP_DocNum에 저장 + 상태 Sent.

DB Status: Draft → Sent

5

SCHEDULED poll · PO creation

Every 1 hour, polls SAP B1 to check if any Sent PR has been converted to PO.

1시간마다 SAP B1 폴링하여 Sent PR의 PO 전환 여부 확인.

API `GET /b1s/v1/PurchaseOrders?`
`$filter=PurchaseRequestNum eq ...`

6

SYSTEM update · PO Created

When SAP B1 generates PO from PR, status → Approved (PO). Triggers WH-01 inbound schedule update.

SAP B1에서 PR이 PO로 전환되면 상태 Approved (PO) → WH-01 입고 예정 갱신.

DOWNSTREAM WH_PurchaseOrder INSERT + WH-01 KPI refresh

7

USER click · [🔄 Retry]

Manual retry for persistently failed PRs (after fixing root cause).

지속 실패 PR에 대한 수동 재시도 (근본 원인 해결 후).

RESET Retry counter reset, attempt fresh transmission

05

Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP06-001

SAP Auth Failure

SAP 인증 실패

CAUSE	Service Layer token expired or invalid.
DETECT	401 response.
ACTION	Auto re-login. Retry. If still fails, alert SYS.
RECOVER	SYS updates credential.

— AUTH FAILED

E-PP06-002

PR Validation Reject

PR 검증 거부

CAUSE	SAP B1 rejects PR (invalid vendor, item, etc.).
DETECT	400 response with reason.
ACTION	Mark Failed with SAP error message. Skip retries.
RECOVER	User fixes data in PP-06 or MD masters.

— SAP REJECTED

E-PP06-003

Network Timeout

네트워크 타임아웃

CAUSE	Service Layer response > 30s.
DETECT	Heartbeat.
ACTION	Retry 3× per BR-PP-006.
RECOVER	Check SAP B1 health.

⚠ TIMEOUT · RETRYING

E-PP06-004

Duplicate PR

PR 중복

CAUSE	Same item+date already has open PR.
DETECT	Pre-send check.
ACTION	Show duplicate, ask user: send anyway or merge.
RECOVER	Merge or cancel duplicate.

⚠️ DUPLICATE

E-PP06-005

Insufficient Vendor

거래선 부재

CAUSE	Item has no default vendor in MD-07.
DETECT	Pre-send.
ACTION	Block send. Cannot create PR without vendor.
RECOVER	MD team sets default vendor in MD-07.

🚫 NO VENDOR

E-PP06-006

SAP B1 Down

SAP 다운

CAUSE	Repeated 5xx errors from SAP.
DETECT	Status check.
ACTION	Halt all sends. Banner: 'SAP B1 unavailable'.
RECOVER	Wait for SAP recovery, then resume.

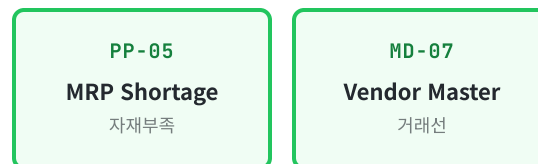
🚫 SAP DOWN

06

Linked Screens & Data Flow

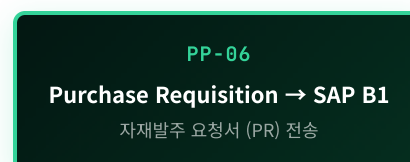
연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



PP-05
MRP Shortage
자재부족

MD-07
Vendor Master
거래선



PP-06
Purchase Requisition → SAP B1
자재발주 요청서 (PR) 전송



SAP B1
PO Creation
구매오더

WH-01
Inbound Schedule
입고예정

▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

07

Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-006	Retry SAP B1 PR transmission 3× on failure; escalate after.	SAP B1 PR 전송 실패 시 3회 재시도, 지속 시 PLAN 알림.	AUT

A-MES · VOL.4 · PP-06 Purchase
Requisition → SAP B1 · L3 Detail Spec
v1.0 · Bilingual EN/KO

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VOL.4 · PP MODULE

PRODUCTION PLAN

생산계획

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PP-02SAP B1 Import공급계획 수신

PP-03Plan & Confirm생산계획 수립

PP-04Work Order Mgmt작업지시 관리

PP-05MRP자재 소요 계획

PP-06Purchase Req.자재발주 요청

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VOL.4 · PP · PRODUCTION PLANNING · WO 발행 및 생산 착수

PP-07

WORK ORDER RELEASE

W O R E L E A S E

DEVICEWeb PC

PHASE1st Dev

ROLEPLAN Manager

FREQDaily · 일간

URL/pp/wo/release

01Overview

화면 개요 · 목적 · 컨텍스트

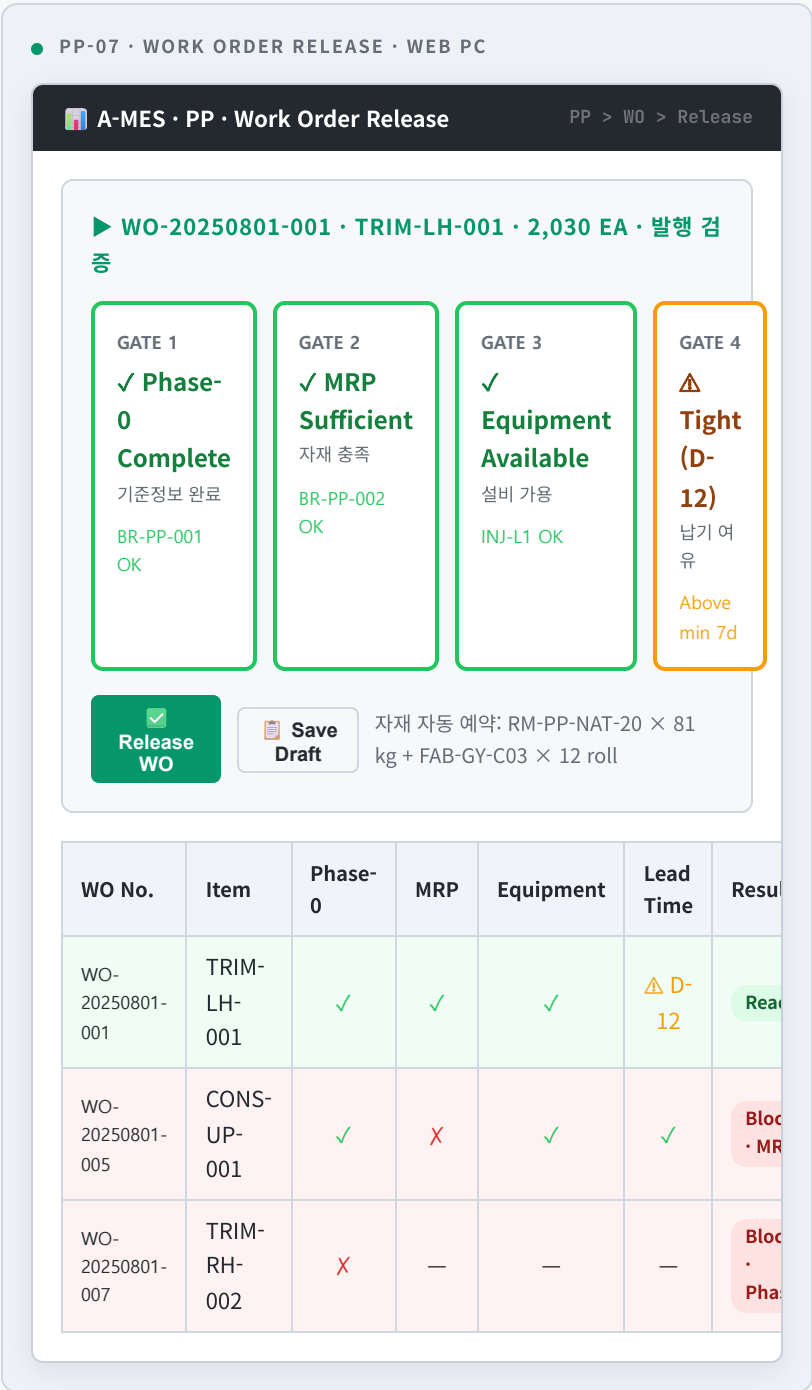
Validates all 4 release conditions sequentially before transitioning WO to Released status: (1) Phase-0 complete, (2) MRP no shortage, (3) Equipment available, (4) Delivery lead time sufficient. On release, materials are auto-reserved (WH_Inventory.Reserved += qty) and the WO becomes visible on INJ/IMG/PNT POP screens. Cancellation auto-releases reservations.

WO Released 전환 전 4가지 조건 순차 검증: (1) Phase-0 완료 · (2) MRP 자재 충족 · (3) 설비 가용 · (4) 납기 여유. 발행 시 자재 자동 예약 (WH_Inventory.Reserved += qty), INJ/IMG/PNT POP 화면 표시. 취소 시 예약 자동 해제.

SCREEN ID	PP-07
GATES	4 sequential validations
EFFECT	Material reservation + POP visibility
CANCEL	Auto-release reservation (BR-PP-004)
TABLES	PP_WO , WH_Inventory

02 Screen Mockup

화면 목업 - 주요 UI 구성



ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증·동작	DEFAULT
CMP-01	Gate 1 · Phase-0 기준정보	AUTO	Y	Checks BOM, BOP, Inspection Standard exist. BR-PP-001 .	—
CMP-02	Gate 2 · MRP MRP 충족	AUTO	Y	Checks WO.IsBlocked = 0 from latest MRP run.	—
CMP-03	Gate 3 · Equipment 설비 가용	AUTO	Y	Checks MNT_Equipment.Status ≠ 'Blocked' for routing equipment.	—
CMP-04	Gate 4 · Lead Time 납기 여유	AUTO	Y	Days until due ≥ min lead time per BOP . Min 7 days default.	—
CMP-05	[☒ Release WO] WO 발행	BUTTON	Y	Enabled only when all 4 gates pass. Updates Status=Released + reserves materials.	—
CMP-06	[Save Draft] Draft 저장	BUTTON	N	Keeps WO in Draft for later release.	—
CMP-07	Reservation Preview 예약 미리보기	DISPLAY	—	Shows materials and qty that will be reserved on release.	—

04 Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER page.load · select WO

User opens release screen for a specific Draft WO.
System runs all 4 gate checks in parallel.

Draft WO 발행 화면 진입. 4개 게이트를 병렬 검증.

API GET /api/pp/wo/:id/release-check

2

SYSTEM validate · Gate 1 Phase-0

Joins WO.item_code with MD-02 (BOM), MD-04 (BOP), MD-06 (Inspection). All must exist.

MD-02/04/06 존재 여부 검증.

RESULT PASS → green check · FAIL → block with specific missing master

3

SYSTEM validate · Gate 2 MRP

Checks WO.IsBlocked set by last MRP. Also re-runs partial MRP for this WO if > 24h since last full.

MRP 실행 후 WO.IsBlocked 확인 + 24시간 경과 시 부분 재실행.

EFFECT BR-PP-002 · auto-trigger partial MRP if stale

4

SYSTEM validate · Gate 3 Equipment

For each routing step in BOP, checks MNT_Equipment availability for the expected start period.

BOP 라우팅별 시작 시점의 설비 가용성 확인.

QUERY `SELECT status FROM mnt_equipment WHERE line_id IN (...) AND start_time BETWEEN ...`

5

SYSTEM validate · Gate 4 Lead

DueDate - TODAY ≥ MinLeadDays (from BOP). Default 7 days.

납기 - 오늘 ≥ 최소 리드(BOP 정의). 기본 7일.

RESULT PASS (green) / WARN (≥3, <7, yellow) / FAIL (<3, red)

6

USER click · [Release WO]

All gates pass (or user accepts warnings). Transitions Status to Released.

전체 게이트 통과 (또는 경고 수락) 시 Status를 Released로 전환.

API `POST /api/pp/wo/:id/release`

7

SYSTEM complete · reserve + visible

Materials auto-reserved in WH_Inventory. WO becomes visible on INJ/IMG/PNT POP. Calendar updates.

자재 자동 예약 + INJ/IMG/PNT POP 표시. 캘린더 갱신.

DOWNSTREAM BR-PP-003 · WH.Reserved += qty · POP visibility · PP-CAL refresh

05

Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP07-001

Gate 1 Failure

기준정보 미완료

CAUSE	Phase-0 incomplete.
DETECT	Gate check.
ACTION	Block release. Show which master is missing.
RECOVER	MD team completes.

⊘ GATE 1 · PHASE-0

E-PP07-002

Gate 2 Failure

MRP 차단

CAUSE	Material shortage from MRP.
DETECT	Gate check.
ACTION	Block. Link to PP-06 for PR.
RECOVER	Generate PR, wait for inbound.

⊘ GATE 2 · MRP

E-PP07-003

Gate 3 Warning

설비 경고

CAUSE	Equipment scheduled for maintenance during WO period.
DETECT	Gate check.
ACTION	Warning popup. Allow with manager approval.
RECOVER	Approve and adjust schedule, or postpone.

⚠ GATE 3 · MNT

E-PP07-004

Gate 4 Tight

납기 임박

CAUSE	Lead time < 7 days.
DETECT	Gate check.
ACTION	Warning. Release allowed but flagged.
RECOVER	Expedite or adjust due date.

E-PP07-005

Reservation Conflict

예약 충돌

CAUSE	Material already fully reserved by other WOs.
DETECT	Reservation transaction.
ACTION	Block. Reservation cannot exceed AvailableQty.
RECOVER	Wait for inbound or cancel other WO.

⊖ RESERVATION

E-PP07-006

Release Permission

발행 권한 없음

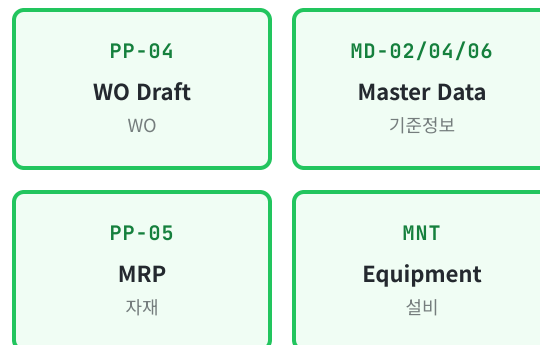
CAUSE	User lacks PP.WO.Release role.
DETECT	RBAC.
ACTION	Block. Only managers can release.
RECOVER	Manager releases or assigns role.

⊖ PERMISSION

06 Linked Screens & Data Flow

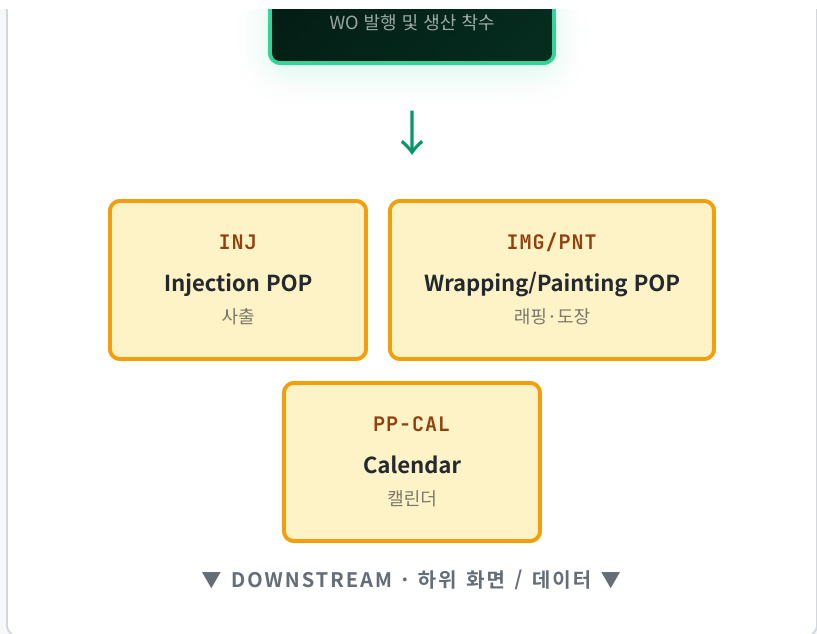
연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



PP-07

Work Order Release



07 Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-001	Block WO creation if Phase-0 master data is incomplete.	Phase-0 기준정보 미완료 시 WO 생성 차단.	BLK
BR-PP-002	Auto-block WO when MRP detects material shortage.	MRP 자재 부족 시 WO 자동 차단.	BLK
BR-PP-003	Auto-reserve materials on WO release.	WO 발행 시 자재 자동 예약.	AUT
BR-PP-007	Warn if equipment is blocked when releasing WO.	WO 발행 시 설비 차단 상태이면 경고.	ALT

VOL.4 · PP · PRODUCTION PLANNING · 생산 캘린더

PP-CAL

PRODUCTION CALENDAR

C A L E N D A R

DEVICE

Web PC

PHASE

2nd Dev

ROLE

PLAN Manager · Viewer

FREQ

Real-time view 실시간 보기

URL

/pp/calendar

01

Overview

화면 개요 · 목적 · 컨텍스트

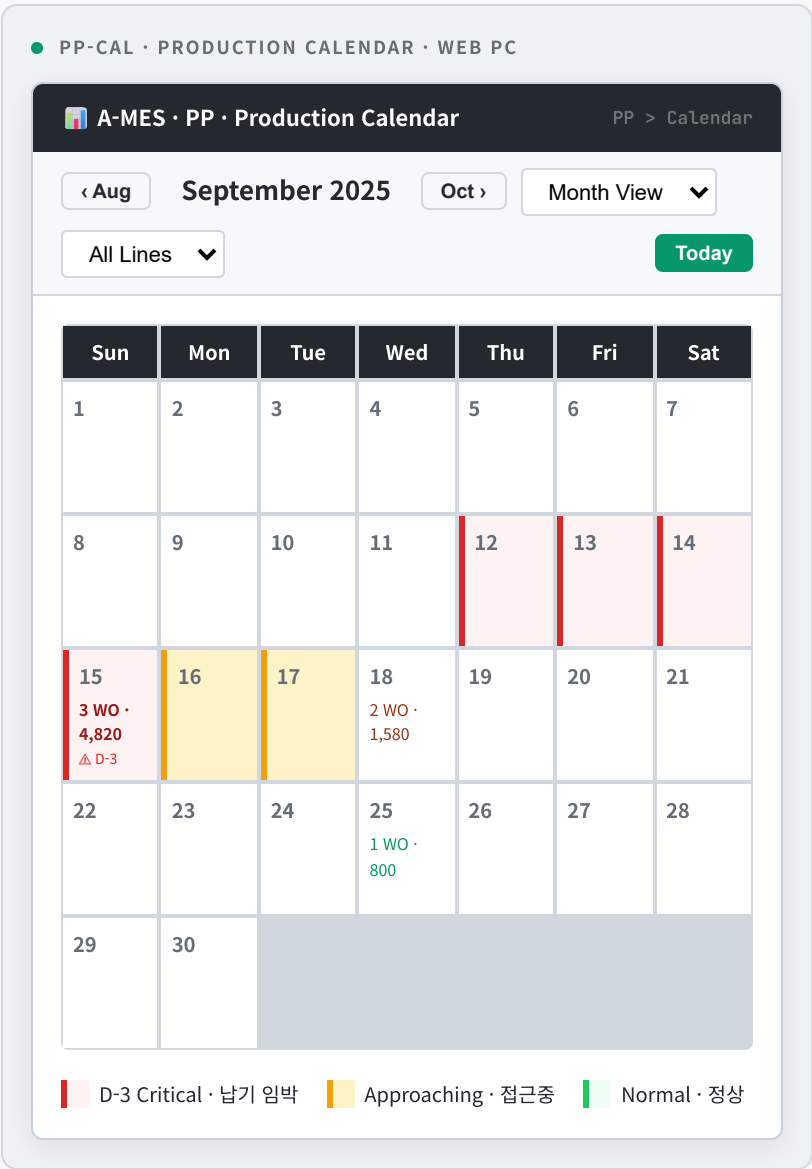
Calendar view of WO due dates with line capacity overlay. Each day shows scheduled WO count, total qty, and capacity utilization. D-3 critical items highlighted red, due-today in orange. Drag-drop to reschedule WO due date (with audit log + manager approval if past today).

WO 납기일을 캘린더 뷰로 표시. 일별로 예정 WO 수·총 수량·능력 가동률 표시. D-3 이내 빨강, 금일 납기 주황 강조. 드래그-드롭으로 납기 재조정(감사 로그 + 과거 일자는 매니저 승인).

SCREEN ID	PP-CAL
VIEW	Month / Week / Day
D-DAY ALERT	D-3 red, today orange (BR-PP-008)
RESCHEDULE	Drag-drop + audit log
TABLES	PP_WO , MD_Equipment (capacity)
RBAC	PP.WO

02

Screen Mockup



03 UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증·동작	DEFAULT
CMP-01	View Mode 보기 모드	SELECT	N	Month / Week / Day. Affects grid density.	Month
CMP-02	Line Filter 라인 필터	SELECT	N	Filter calendar by production line.	ALL

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증·동작	DEFAULT
CMP-03	Today Button 오늘	BUTTON	N	Jumps to today's date.	—
CMP-04	Day Cell 일자 셀	INTERACT	—	Shows WO count, total qty, capacity %. Click expands list. Drag-drop reschedules.	—
CMP-05	WO Drag-Drop WO 드래그	INTERACT	N	Drag WO chip to another day to reschedule. Past date requires manager approval.	—
CMP-06	D-Day Color Code D-Day 색상	AUTO	—	Red ≤ D-3, orange today, green normal. BR-PP-008 .	—
CMP-07	Capacity Overlay 능력 오버레이	DISPLAY	—	Per-day load %. Red > 90%, yellow 70-90%.	—

Event & Action

04

Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER page.load · open calendar

Renders current month with all WOs by due date.
Capacity overlay loaded from MNT.

현재 월에 WO 표시 + MNT 능력 오버레이.

API GET /api/pp/calendar?month=YYYY-MM

2

USER click · day cell

Expands the day to show full WO list with item, qty, status.

일자 셀 클릭 시 전체 WO 목록(품목·수량·상태) 표시.

UI Modal or expand-row

3

USER drag · WO to new date

User drags WO chip to another day. System validates: not past today, capacity available, lead time OK.

WO를 다른 일자로 드래그. 과거 일자·능력·리드타임 검증.

VALIDATE Block past date without approval. Capacity check.

4

SYSTEM save · reschedule

Updates PP_WO.due_date with audit log entry.

PP_WO.due_date 갱신 + 감사 로그.

AUDIT Audit: old_date, new_date, user, reason

5

SYSTEM alert · D-3 daily check

Every day at 00:00, scans all WOs and flags those reaching D-3 today.

매일 00:00 D-3 도달 WO 자동 검색·플래그.

ALERT BR-PP-008 · email + UI highlight

6

SYSTEM sync · downstream

Reschedule propagates to PP-04 list and PP-OTD KPI projection.

재조정이 PP-04 목록·PP-OTD KPI 예측에 반영.

DOWNSTREAM PP-04 / PP-OTD refresh

05

Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PPCAL-001

Past Date Drop

과거 일자 드롭

CAUSE	User drops WO to a past date.
DETECT	Drop validation.
ACTION	Block. Modal: 'Cannot reschedule to past. Approve as manager?'
RECOVER	Manager override with reason.

⚠ PAST DATE

E-PPCAL-002

Capacity Exceeded

능력 초과

CAUSE	New date capacity > 100%.
DETECT	Capacity check.
ACTION	Yellow warning. Allow with note.
RECOVER	Confirm with capacity manager.

⚠ OVER CAPACITY

E-PPCAL-003

Lead Time Violation

리드타임 위반

CAUSE	New date doesn't allow required production lead.
DETECT	Lead check.
ACTION	Block. Show min required date.
RECOVER	Choose later date.

⊖ LEAD INSUFFICIENT

E-PPCAL-004

Closed WO Drag

마감 WO 드래그

CAUSE	User drags a Closed WO.
DETECT	Status check.
ACTION	Block. Closed WOs are immutable.
RECOVER	—

⊖ CLOSED

E-PPCAL-005

Calendar Load Slow

캘린더 로딩 지연

CAUSE	Many WOs in month, render slow.
DETECT	Render time monitor.
ACTION	Show loading spinner. Virtualize cells.
RECOVER	Narrow filter.

⚠ SLOW LOAD

E-PPCAL-006

Permission to Reschedule

재조정 권한

CAUSE User lacks PP.WO.Edit role.

DETECT RBAC.

ACTION Drag disabled. View-only mode.

RECOVER Request role.

🚫 READ ONLY

06

Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲

PP-04

WO Master

WO

MNT

Equipment Capacity

능력



PP-CAL

Production Calendar

생산 캘린더



PP-OTD

OTD KPI

납기

PP-04

WO Updates

WO 갱신

▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

07

Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-008	Alert PLAN/MGR when WO due date is within D-3.	WO 납기 D-3 이내 강조 알 림.	ALT

A-MES · VOL.4 · PP-CAL Production
Calendar · L3 Detail Spec v1.0 ·
Bilingual EN/KO

Automotive Parts MES ·
SAV & GEO Supply ·
Confidential

A-MES

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WH Warehouse

PP Plan

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PP-OTD · On-Time Delivery KPI

VOL.4 · PP MODULE

PRODUCTION PLAN

생산계획

L3 SCREENS · 화면 상세

PP-01

Forecast Mgmt

Forecast 관리

PP-02

SAP B1 Import

공급계획 수립

PP-03

Plan & Confirm

생산계획 수립

PP-04

Work Order Mgmt

작업지시 관리

PP-05

MRP

자재 소요 계획

PP-06

Purchase Req.

자재발주 요청

PP-07

WO Release

WO 발행

PP-CAL

Production Calendar

생산 캘린더

PP-OTD

Delivery KPI

납기 현황

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VOL.4 · PP · PRODUCTION PLANNING · 납기 준수율 현황

PP-OTD

ON-TIME DELIVERY KPI

O T D K P I

DEVICE

Web PC

PHASE

2nd Dev

ROLE

PLAN Manager · Director

FREQ

Real-time + daily summary

URL

/pp/otd

01

Overview

화면 개요 · 목적 · 컨텍스트

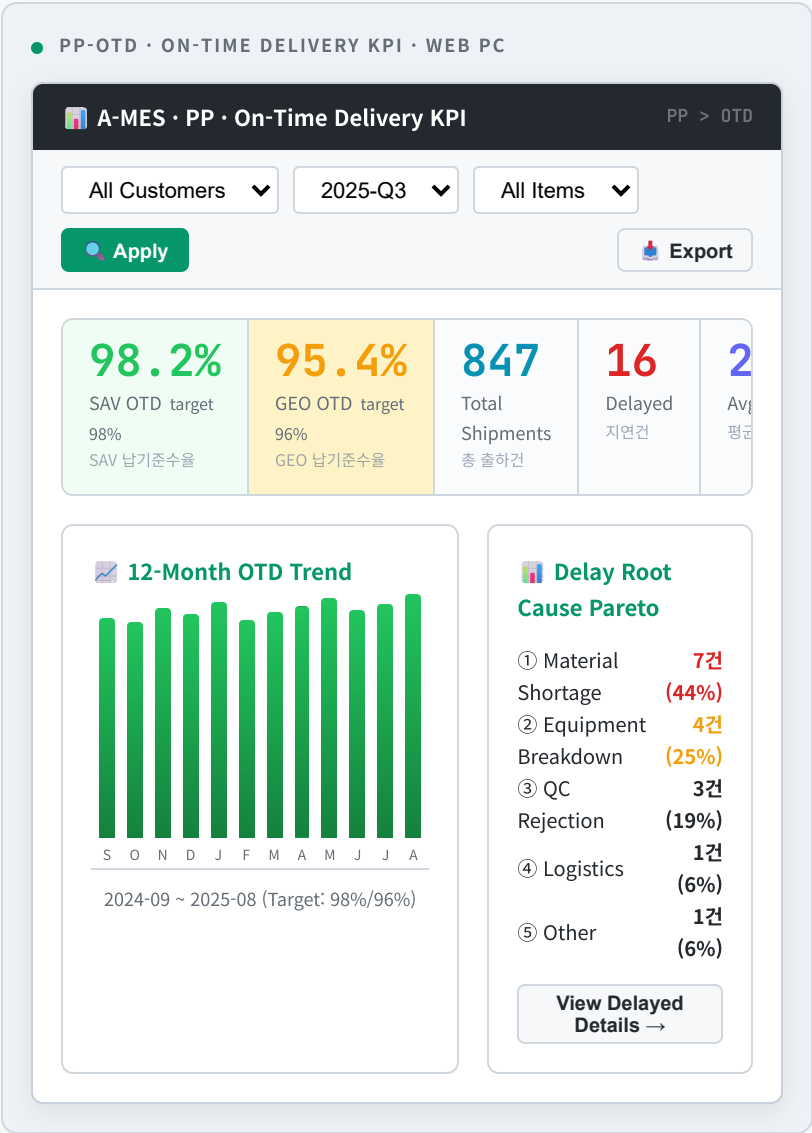
Tracks On-Time Delivery (OTD) KPI by customer (SAV/GEO), item, and period. Calculates $OTD\% = (\text{Shipped on or before due} / \text{Total shipped}) \times 100$. Targets: SAV $\geq 98\%$, GEO $\geq 96\%$. Shows trend chart (12-month), delay root-cause Pareto, and individual delayed shipment drill-down. Feeds RPT-05 monthly KPI dashboard.

거래선(SAV/GEO)·품목·기간별 OTD KPI 추적. $OTD\% = (\text{납기 내 출하} / \text{총 출하}) \times 100$. 목표: SAV $\geq 98\%$ · GEO $\geq 96\%$. 12개월 추이 차트·지연 원인 Pareto·개별 지연건 드릴다운. RPT-05 월간 KPI 대시보드 연계.

SCREEN ID	PP-OTD
TARGET	SAV $\geq 98\%$ · GEO $\geq 96\%$
FORMULA	$(\text{On-time} / \text{Total}) \times 100$
SOURCE	FG_Shipment.actual_date vs PP_WO.due_date
TABLES	FG_Shipment , PP_WO
RBAC	PP.View · Director

02 Screen Mockup

화면 목업 — 주요 UI 구성



03 UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-01	Customer Filter 고객사	SELECT	N	All / SAV / GEO.	All
CMP-02	Period Filter 기간	SELECT	N	Q/M/Custom date range.	Current Q

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-03	Item Filter 품목	SELECT	N	Filter by specific item.	All
CMP-04	OTD Card OTD 카드	DISPLAY	—	OTD% = (On-time / Total) × 100 . Target indicator.	—
CMP-05	Trend Chart 추이 차트	CHART	—	12-month bar chart. Click bar drills into that month.	—
CMP-06	Pareto Chart Pareto	CHART	—	Top 5 delay reasons. Click row drills to detailed cases.	—
CMP-07	[View Delayed Details] 지연 상세	BUTTON	N	Drill into individual delayed shipments with root cause.	—

04 Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER page.load · open OTD

Loads current period OTD with all filters at default. Calls aggregate API.

현재 기간 OTD를 기본 필터로 로드. 집계 API 호출.

API GET /api/pp/otd?period=current_quarter

2

SYSTEM calc · OTD%

Aggregates: FG_Shipment joined with PP_WO. Counts where actual_ship_date ≤ wo_due_date.

FG_Shipment × PP_WO 조인. actual_ship_date ≤ wo_due_date 건수 집계.

3

FORMULA `OTD% = COUNT(on_time) / COUNT(*) * 100`

USER `click · trend bar`

Drills into selected month. Re-fetches detailed shipments for that month.

추이 차트 막대 클릭 시 해당 월 상세 출하 데이터 조회.

API `GET /api/pp/otd/detail?month=YYYY-MM`

4

USER `click · Pareto row`

Drills into shipments with that delay reason. Shows root cause analysis link to QC/MNT data.

Pareto 행 클릭 시 해당 사유 지연건 목록 + QC/MNT 데이터 링크.

NAV `/pp/otd/detail?cause=MaterialShortage`

5

SCHEDULED `cron · daily KPI snapshot`

Every day at 00:30, snapshots OTD for previous day into PP_OTD_Snapshot.

매일 00:30 전일 OTD 스냅샷을 PP_OTD_Snapshot에 저장.

EFFECT Used for fast trend rendering

6

SYSTEM `feed · monthly KPI`

Aggregated to RPT-05 monthly KPI dashboard. Director-level visibility.

월간 RPT-05 KPI 대시보드에 집계 전달. 임원진 가시성.

DOWNSTREAM RPT-05 · Email digest to plant director on miss

05

Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP0TD-001

No Shipments

출하 없음

CAUSE	No FG shipments in filter range.
DETECT	Empty data.
ACTION	Show empty state.
RECOVER	Widen date range.

i NO DATA

E-PP0TD-002

Below Target

목표 미달

CAUSE	OTD < target (SAV 98% / GEO 96%).
DETECT	Daily check.
ACTION	Red KPI card. Alert plant director.
RECOVER	Root cause analysis + CAPA.

⚠ BELOW TARGET

E-PP0TD-003

Missing WO Reference

WO 참조 누락

CAUSE	Shipment has no linked WO (manual ship?).
DETECT	Join NULL.
ACTION	Excluded from OTD calc. Listed separately.
RECOVER	Reconcile in FG module.

⚠ ORPHAN SHIPMENT

E-PP0TD-004

Slow Query

쿼리 지연

CAUSE	Aggregation > 10s for large period.
DETECT	Timeout monitor.
ACTION	Use snapshot table or narrow scope.
RECOVER	Apply more specific filter.

⚠ SLOW · USE SNAPSHOT

E-PP0TD-005

Snapshot Missing

스냅샷 결손

CAUSE	Daily snapshot job failed yesterday.
DETECT	Existence check.
ACTION	Re-aggregate from raw. Mark with warning badge.
RECOVER	SYS reruns snapshot job.

⚠ NO SNAPSHOT

E-PP0TD-006

Export Permission

출력 권한 없음

CAUSE	User lacks PP.OTD.Export role.
DETECT	RBAC.
ACTION	Block export. View only.
RECOVER	Request role.

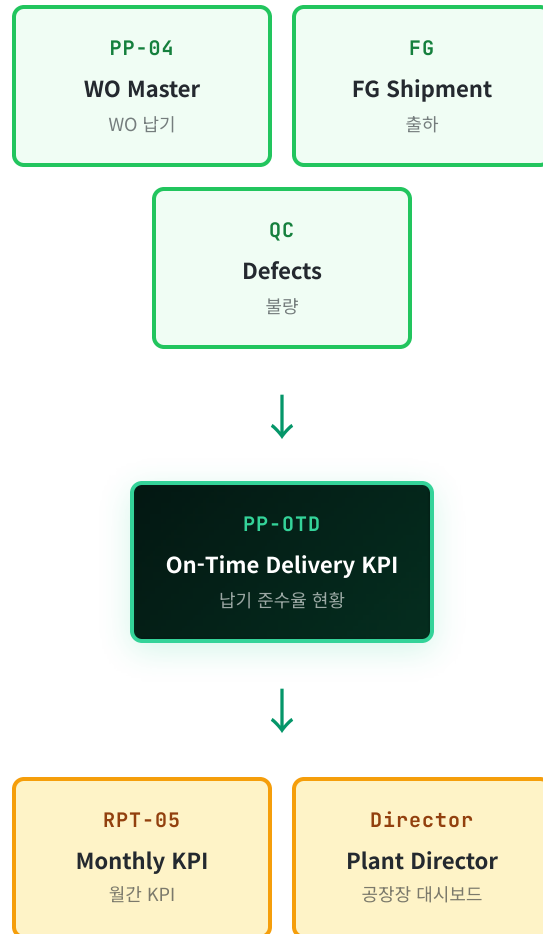
🚫 EXPORT DENIED

06

Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

RULE ID	RULE (EN)	규칙 (KO)	ACTION
No specific business rules apply to this screen. 이 화면에 직접 적용되는 업무 규칙은 없습니다.			

A-MES · VOL.4 · PP-OTD On-Time DeliveryAutomotive Parts MES ·
KPI · L3 Detail Spec v1.0 · Bilingual SAV & GEO Supply ·
EN/KO Confidential

A-MESIndex

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WO 발행

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Production Calendar

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PP-0TD

Delivery KPI

납기 현황

LINE SCHEDULING · 라인 스케줄링

PP-LSB

Line Schedule Board

라인 스케줄 보드

PP-0DM

Operating/Downtime Monitor

가동·비가동 모니터

PP-OEE

Line Availability & OEE

라인 가용성·OEE

PP-DTL

Downtime Log Entry

비가동 로그 입력

MD-29

Line Time Pattern

라인 시간패턴

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03

UI Components

VOL.4 · PP · LINE SCHEDULING · 라인 스케줄 보드

PP-LSB

LINE SCHEDULE BOARD

라인 별 분 단 위 스 케 줄 보 드

DEVICE

Web PC

PHASE

2nd Dev

ROLE

PLAN Manager

FREQ

Daily · 일간

URL

/pp/line-schedule

01

Overview

화면 개요 · 목적 · 컨텍스트

A minute-level Gantt board that schedules Work Orders onto each line's **own time pattern** (MD-29). Every line row renders its active MD_LineTimePattern as background bands — green OPERATING windows and amber PLANNED_DOWN windows — and WO blocks (routing A blue / B orange) are dragged onto operating windows only. Available capacity per line = Σ OPERATING minutes; the board blocks any WO that would overflow an operating window or land on planned downtime.

작업지시(WO)를 각 라인의 **고유 시간패턴**(MD-29) 위에 배치하는 분단위 간트 보드. 각 라인 행은 활성 MD_LineTimePattern을 배경 밴드로 렌더링한다 — 녹색 OPERATING 시간창, 호박색 PLANNED_DOWN 시간창 — 그리고 WO 블록 (라우팅 A 파랑/B 주황)은 가동 시간창에만 드래그 배치된다. 라인별 가용 캐파 = Σ OPERATING 분; 가동창을 초과하거나 계획 비가동에 걸치는 WO는 차단된다.

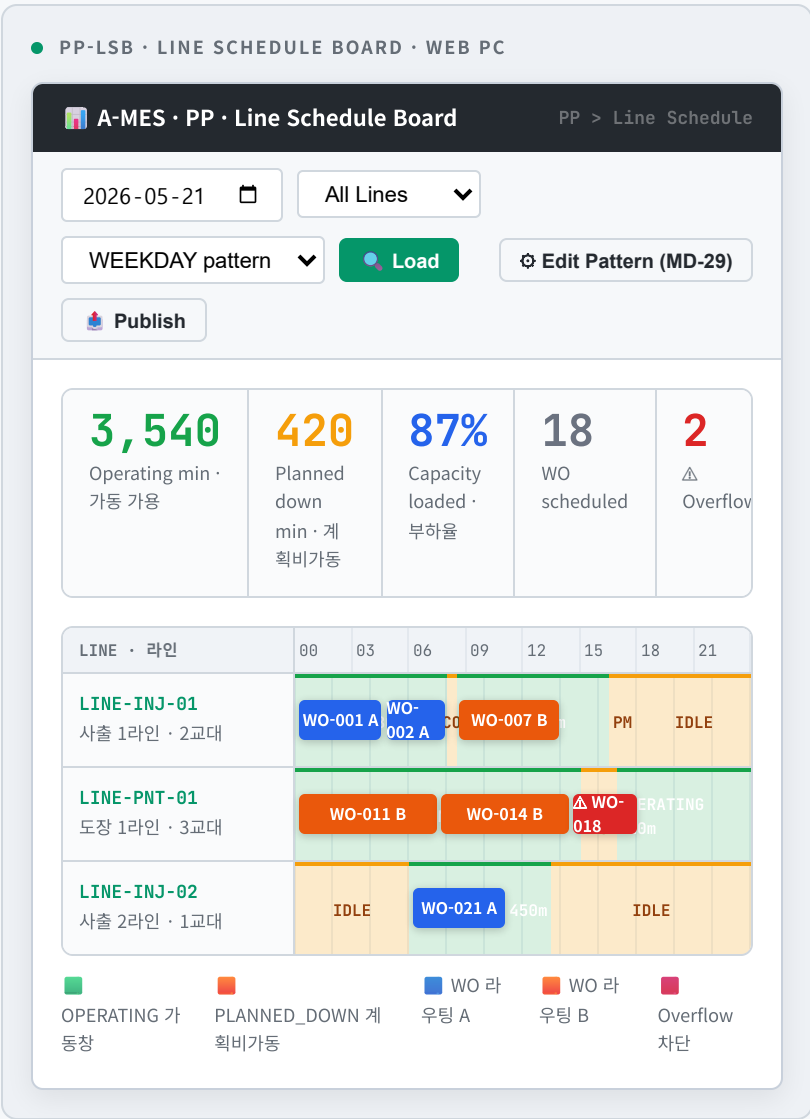
SCREEN ID	PP-LSB
GRANULARITY	1-minute slot · 분단위
BASELINE	MD-29 Line Time Pattern

TABLES	MD_LineTimePattern , MD_LineTimeSegment , PP_LineSchedule , PP_WO
RBAC	PP.LineSchedule
DOWNSTREAM	PP-ODM · PP-OEE · INJ/PNT POP

02

Screen Mockup

화면 목업 — 분단위 간트 보드



03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-01	Date Picker 대상 일자	DATE	Y	Selected date resolves each line's pattern by DayType (WEEKDAY/SAT/HOLIDAY) via MD-14 Calendar.	Today
CMP-02	Line Filter 라인 필터	SELECT	N	All / INJ lines / PNT lines / single line.	All
CMP-03	Pattern Band 시간패턴 밴드	AUTO	—	Renders MD_LineTimeSegment rows: OPERATING green, PLANNED_DOWN amber. Minute-positioned via <code>StartMin/EndMin</code> .	—
CMP-04	WO Block (drag) WO 블록	DRAG	—	Drag a WO onto an OPERATING window. Snap to 1-min grid. Drop on PLANNED_DOWN or overflow → rejected (E-LSB-001).	—
CMP-05	Capacity Bar 부하율 바	AUTO	—	$\text{Load\%} = \frac{\sum \text{WO minutes}}{\sum \text{OPERATING minutes}}$. Red >100%, amber 85-100%, green <85%.	—
CMP-06	Edit Pattern 패턴 편집	NAV	—	Opens MD-29 pattern editor in modal. On save, board reloads bands. Requires <code>MD.LineTimePattern</code> role.	—
CMP-07	Publish 발행	BUTTON	N	Commits PP_LineSchedule rows; pushes schedule to INJ/PNT POP and PP-ODM baseline. Blocked if any overflow remains.	—
CMP-08	Auto-Schedule 자동 배치	BUTTON	N	Heuristic fill of released WOs into OPERATING windows by D-Day ascending; user reviews before Publish.	—

1

USER `page.load` · `select date`

Resolves each line's DayType from MD-14 Calendar, loads the matching ACTIVE MD_LineTimePattern and renders minute-level bands.

MD-14 캘린더로 라인별 요일유형 판별 → ACTIVE 시간패턴 로드
→ 분단위 밴드 렌더링.

API `GET /api/pp/line-schedule?date=2026-05-21`

2

USER `drag` · `WO block` → `operating window`

User drags a released WO onto an OPERATING segment; system snaps to 1-minute grid and computes end minute from WO standard run time.

발행 WO를 OPERATING 세그먼트에 드래그 → 1분 그리드 스냅
→ WO 표준시간으로 종료 분 계산.

CALC $\text{endMin} = \text{startMin} + \text{ceil}(\text{WO.qty} \times \text{cycleTime} / 60)$

3

SYSTEM `validate` · `window fit`

Checks the WO block stays inside one OPERATING window and does not cross a PLANNED_DOWN segment; overflow flips the block red.

WO 블록이 단일 OPERATING 창 내에 있고 PLANNED_DOWN을 가로지르지 않는지 검증 — 초과 시 빨강 표시.

RULE BR-PP-031 · overflow → block + E-LSB-001

4

USER `click` · `[Auto-Schedule]`

Heuristically fills all unscheduled released WOs into available OPERATING minutes, ordered by D-Day ascending.

미배치 발행 WO를 D-Day 오름차순으로 가용 OPERATING 분에 자동 충전.

API `POST /api/pp/line-schedule/auto-fill`

5

USER click · [Publish]

Commits PP_LineSchedule, pushes the day's schedule to INJ/PNT POP screens and sets the PP-ODM operating baseline.

PP_LineSchedule 확정 → INJ/PNT POP에 일정 푸시 → PP-ODM 가동 기준선 설정.

DOWNSTREAM PP-ODM baseline · PP-OEE plan · POP work queue

05

Exception Handling

예외 처리 · 오류 코드

E-LSB-001

Window Overflow

가동창 초과

CAUSE	WO block exceeds the OPERATING window or crosses a PLANNED_DOWN segment.
DETECT	On drop / auto-fill.
ACTION	Block turns red, Publish disabled. Split WO or move to another window.

OVERFLOW · BLOCKED

E-LSB-002

No Active Pattern

패턴 미정의

CAUSE	Line has no ACTIVE MD_LineTimePattern for the resolved DayType.
DETECT	On page load.
ACTION	Line row greyed; prompt to define pattern in MD-29.

PATTERN MISSING

E-LSB-003

Capacity Over 100%

부하율 초과

CAUSE	Σ WO minutes exceeds Σ OPERATING minutes for a line.
DETECT	On every schedule change.
ACTION	Capacity bar red; suggest moving WO to another line / day.

⚠ OVER-CAPACITY

E-LSB-004

Pattern Changed After Publish

발행 후 패턴 변경

CAUSE	MD-29 pattern edited after the day was published.
DETECT	Pattern version compare.
ACTION	Banner: re-validate & re-publish required.

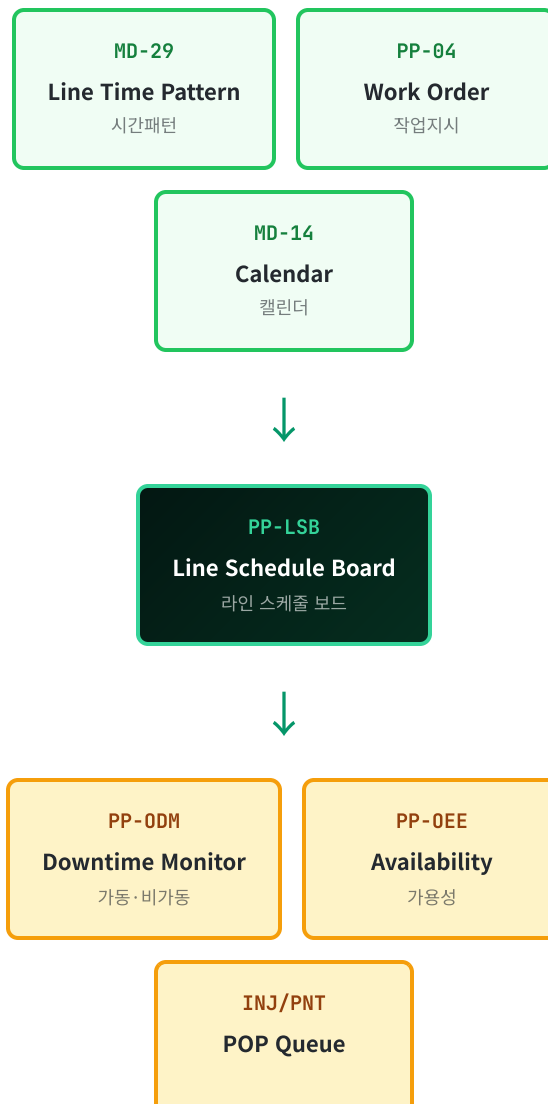
⚠ STALE SCHEDULE

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Linked Screens & Data Flow

연계 화면 · 상·하류 흐름

▲ UPSTREAM ▲



07

Applicable
Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-030	Each line schedules on its own MD-29 time pattern, not the shared factory calendar.	각 라인은 공용 캘린더가 아닌 고유 MD-29 시간패턴으로 스케줄링.	AUT
BR-PP-031	A WO block must fit entirely within one OPERATING window.	WO 블록은 단일 OPERATING 창 안에 완전히 들어가야 함.	BLK
BR-PP-032	Line capacity = Σ OPERATING minutes; load over 100% blocks Publish.	라인 캐파 = Σ OPERATING 분; 부하율 100% 초과 시 발행 차단.	BLK
BR-PP-033	Publish pushes the schedule to POP and sets the PP-ODM operating baseline.	발행 시 POP에 일정 푸시 + PP-ODM 가동 기준선 설정.	AUT
BR-PP-034	A pattern edited after publish flags the schedule stale for re-validation.	발행 후 패턴 변경 시 일정을 stale 표기 — 재검증 필요.	ALT

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WH WarehousePP PlanINJ InjectionIMG WrappingPNT PaintingQC QualityFG Finished GoodsMNT Main

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VOL.4 · PP MODULE

PRODUCTION PLAN

생산계획

L3 SCREENS · 화면 상세

PP-01Forecast MgmtForecast 관리

PP-02SAP B1 Import공급계획 수신

PP-03Plan & Confirm생산계획 수립

PP-04Work Order Mgmt작업지시 관리

PP-05MRP자재 소요 계획

PP-06Purchase Req.자재발주 요청

PP-07WO ReleaseWO 발행

PP-CALProduction Calendar생산 캘린더

PP-0TDDelivery KPI납기 현황

LINE SCHEDULING · 라인 스케줄링

PP-LSBLine Schedule Board라인 스케줄 보드

PP-ODMOperating/Downtime Monitor가동·비가동 모니터

PP-OEELine Availability & OEE라인 가용성·OEE

PP-DTLDowntime Log Entry비가동 로그 입력

MD-29Line Time Pattern라인 시간패턴

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VOL.4 · PP · LINE SCHEDULING · 가동·비가동 모니터

OPERATING / DOWNTIME MONITOR

가동 · 계획 비가동 · 실제 비가동 모니터

DEVICE Web PC / 현황판PHASE 2nd Dev

ROLE PLAN · MNT · SupervisorFREQ Realtime · 실시간

URL /pp/downtime-monitor

01Overview

화면 개요 · 목적 · 컨텍스트

A real-time wallboard that compares each line's **planned time pattern** (MD-29 baseline) against its **actual operating state** streamed from equipment signals. Every minute of the day resolves to one of three states. When the plan says OPERATING but the line signal says stopped, the gap is auto-classified **UNPLANNED_DOWN** and the operator is prompted to log a reason via PP-DTL. The classified state log feeds PP-OEE availability and RPT-06 OEE.

각 라인의 **계획 시간패턴**(MD-29 기준선)과 설비 신호로 수신되는 **실제 가동 상태**를 실시간 대조하는 현황판. 하루의 매 분(分)이 세 상태 중 하나로 판정된다. 계획상 OPERATING인데 라인 신호가 정지되면 그 구간은 **UNPLANNED_DOWN**으로 자동 분류되고 작업자에게 PP-DTL 사유 입력을 요청한다. 분류된 상태 로고는 PP-OEE 가용성·RPT-06 OEE로 전달된다.

SCREEN ID	PP-ODM
REFRESH	WebSocket · 1-minute tick

BASELINE	MD-29 ACTIVE pattern
TABLES	PP_LineStateLog , MD_LineTimePattern , PP_LineDowntimeLog
RBAC	PP.DowntimeMonitor
DOWNSTREAM	PP-DTL · PP-OEE · RPT-06

OPERATING

가동

계획 OPERATING 창에서 설비 신호가 RUN. 정상 생산 시간으로 집계되어 OEE 가용시간 (Run Time)에 반영.

PLANNED_DOWN

계획 비가동

MD-29 패턴이 PLANNED_DOWN으로 정의한 구간 (PM·금형교체·휴게). OEE 부하시간 (분모)에서 차감되는 정상 손실.

UNPLANNED_DOWN

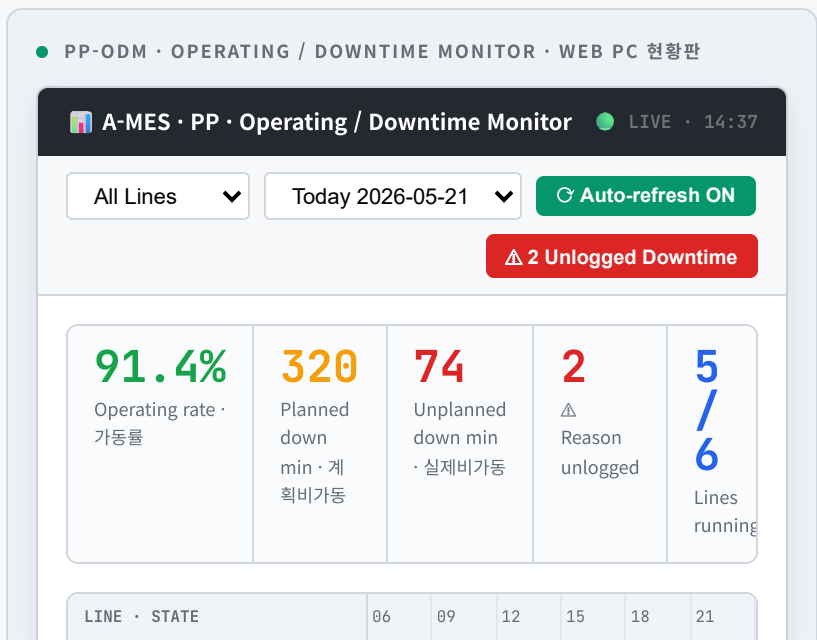
실제 비가동

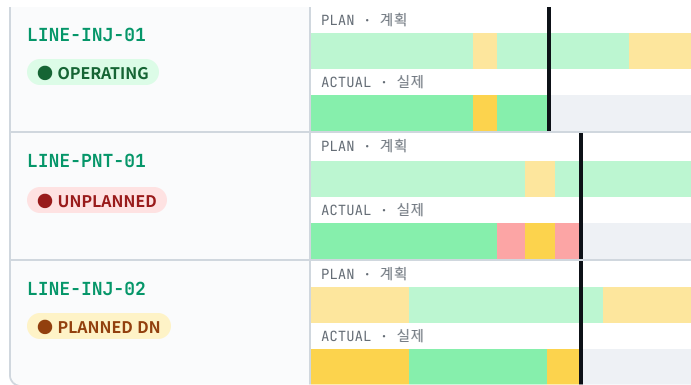
계획상 OPERATING이나 설비 정지. 고장·자재대기·품질정지 등 — PP-DTL 사유 입력 필요, OEE 가용성 손실로 집계.

02

Screen Mockup

화면 목업 — 실시간 가동·비가동 현황판





Time	Line	State	Duration	Reason · 사유	Action
13:58~ 14:09	LINE-PNT-01	UNPLANNED	11 min	△ 미입력	사유 입력
14:24~ 14:37	LINE-PNT-01	UNPLANNED	13 min 🕒	△ 진행중	사유 입력
12:00~ 12:48	LINE-INJ-02	PLANNED	48 min	RSN-DT-PM · 예방보전	정상

03 UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-01	Plan vs Actual Lane 계획 · 실제 레인	AUTO	—	Two stacked minute bands per line — PLAN from MD-29, ACTUAL from PP_LineStateLog. NOW vertical line marks current minute.	—
CMP-02	State Classifier 상태 분류기	AUTO	—	Per minute: signal RUN + plan OPERATING → OPERATING · plan PLANNED_DOWN → PLANNED_DOWN · signal STOP + plan	—

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
				OPERATING → UNPLANNED_DOWN.	
CMP-03	Operating Rate KPI 가동률	AUTO	—	Rate = OPERATING min / (OPERATING + UNPLANNED_DOWN) min elapsed today.	—
CMP-04	Unlogged Downtime Alert 미입력 경보	AUTO	—	Counts UNPLANNED_DOWN events with no PP_LineDowntimeLog reason. Red badge + row highlight.	0
CMP-05	Log Reason Button 사유 입력	NAV	—	Opens PP-DTL pre-filled with line, start/end minute and pattern reference.	—
CMP-06	Line Filter 라인 필터	SELECT	N	All / by process / single line.	All
CMP-07	Auto-refresh 자동 갱신	TOGGLE	N	WebSocket 1-minute tick. When off, manual refresh only.	ON

04 Event & Action Flow

이벤트 흐름 · 상태 분류 → 사유 입력

1

TICK minute.tick · equipment signal

Every minute, equipment RUN/STOP signal (via SYS-04 interface) is paired with the line's MD-29 planned state for that minute.

매 분 설비 RUN/STOP 신호(SYS-04 인터페이스)를 해당 분의 MD-29 계획 상태와 짝지음.

```
IN signal{lineId, minute, runFlag} +
MD_LineTimeSegment.state
```

2

SYSTEM classify · 3-state

Resolves the minute to OPERATING / PLANNED_DOWN / UNPLANNED_DOWN and appends a PP_LineStateLog row.

해당 분을 OPERATING / PLANNED_DOWN /
UNPLANNED_DOWN으로 판정 → PP_LineStateLog 행 추가.

RULE BR-PP-035 · STOP within OPERATING window →
UNPLANNED_DOWN

3

SYSTEM detect · downtime event

Consecutive UNPLANNED_DOWN minutes are merged
into one open downtime event; the board badges it
as reason-unlogged.

연속 UNPLANNED_DOWN 분(分)을 하나의 진행중 비가동 이벤트
로 병합 — 사유 미입력 배지 표시.

API POST /api/pp/downtime-event · status=OPEN

4

SYSTEM alert · downtime threshold

An unplanned downtime exceeding 10 minutes
pushes an alert to PLAN / MNT supervisors.

실제 비가동 10분 초과 시 PLAN/MNT 감독자에게 경보 푸시.

ALERT BR-PP-037 · in-app + SYS-06 notification

5

USER click · [사유 입력] → PP-DTL

Operator logs a downtime reason; the event closes
and the classified state log flows to PP-OEE
availability.

작업자가 비가동 사유 입력 → 이벤트 종료 → 분류 상태 로그가
PP-OEE 가용성으로 전달.

DOWNSTREAM PP-DTL close · PP-OEE recalc · RPT-06 OEE

05

Exception Handling

예외 처리 · 오류 코드

E-ODM-001

Signal Loss

신호 두절

CAUSE	No equipment RUN/STOP signal received for > 3 minutes.
DETECT	Heartbeat timeout.
ACTION	Lane greyed as UNKNOWN; minutes held pending, not classified.

 SIGNAL LOST

E-ODM-002

Reason Unlogged

사유 미입력

CAUSE	UNPLANNED_DOWN event closed without a reason for > 30 min.
DETECT	Scheduled sweep.
ACTION	Escalation alert to supervisor; event stays OPEN.

⚠ REASON OVERDUE

E-ODM-003

Run During Planned Down

계획비가동 중 가동

CAUSE	Equipment RUN while plan says PLANNED_DOWN.
DETECT	Classifier mismatch.
ACTION	Counted as OPERATING (extra run); flagged for pattern review.

⚠ UNPLANNED RUN

E-ODM-004

No Pattern Baseline

기준 패턴 없음

CAUSE	Line has no ACTIVE MD-29 pattern for today.
DETECT	On monitor load.
ACTION	Cannot classify; lane shows 'define pattern' prompt.

🚫 BASELINE MISSING

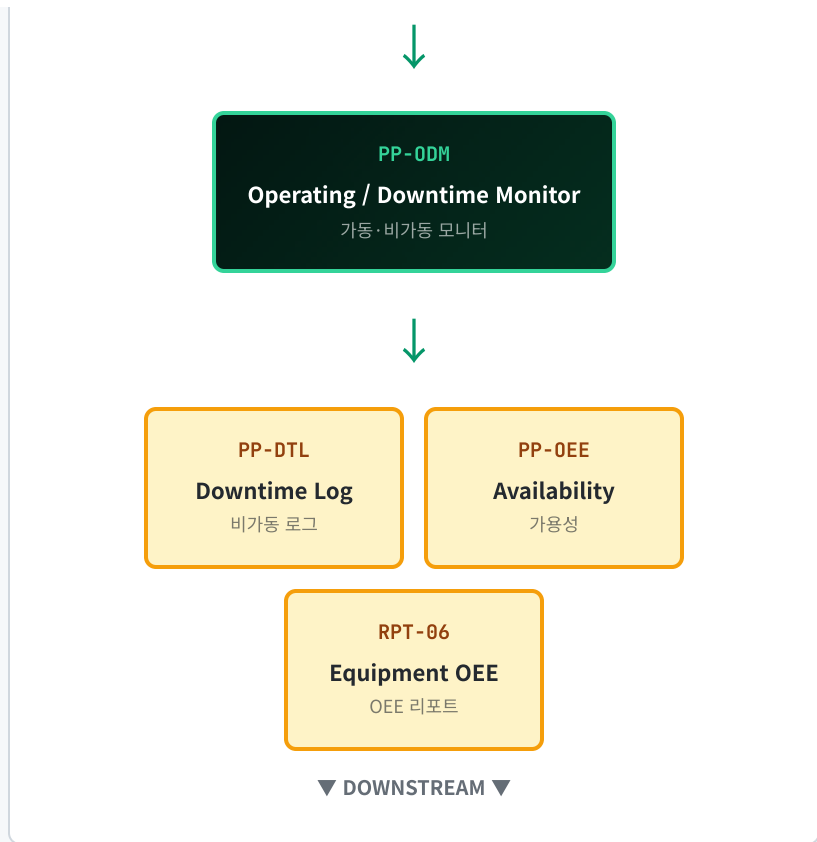
06

Linked Screens & Data Flow

연계 화면 · 상·하류 흐름

▲ UPSTREAM ▲





07 Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-035	Equipment STOP inside a planned OPERATING window is auto-classified UNPLANNED_DOWN.	계획 OPERATING 창 내 설비 정지는 UNPLANNED_DOWN 자동 분류.	AUT
BR-PP-036	Every UNPLANNED_DOWN event requires a reason code logged via PP-DTL.	모든 실제 비가동 이벤트는 PP-DTL 사유코드 입력 필수.	BLK
BR-PP-037	Unplanned downtime over 10 minutes escalates an alert to PLAN / MNT.	실제 비가동 10분 초과 시 PLAN/MNT 경보 에스컬레이션.	ALT
BR-PP-038	Operating rate = OPERATING / (OPERATING + UNPLANNED_DOWN)	가동률 = 가동 / (가동 + 실제비가동) - 계획 비가동 제외.	AUT

RULE ID	RULE (EN)	규칙 (KO)	ACTION
	excludes planned downtime.		
BR-PP-039	The classified PP_LineStateLog is the single source for PP-OEE and RPT-06.	분류된 PP_LineStateLog가 PP-OEE·RPT-06의 단 일 기준 데이터.	AUT

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VOL.4 · PP MODULE

PRODUCTION PLAN

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LINE SCHEDULING · 라인 스케줄링

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MD-29Line Time Pattern라인 시간패턴

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VOL.4 · PP · LINE SCHEDULING · 라인 가용성·OEE

PP-OEELINE AVAILABILITY & OEE

라인 가용성 · OEE 롤업

DEVICEWeb PC

PHASE2nd Dev

ROLEPLAN · MGT

FREQDaily / Period

URL/pp/line-oee

01Overview

화면 개요 · 목적 · 컨텍스트

Rolls up the classified PP_LineStateLog into a time-loss model and computes OEE per line and period. Loading Time (Σ planned OPERATING from MD-29) minus PLANNED_DOWN and UNPLANNED_DOWN yields the Operating Time and **Availability**; combined with Performance (cycle-time efficiency) and Quality (QC yield) it produces OEE. The screen separates planned vs unplanned loss so each line's own time pattern is fairly credited.

분류된 PP_LineStateLog를 시간손실 모델로 롤업하여 라인·기간별 OEE를 산출한다. 부하시간(MD-29 계획 OPERATING 합계)에서 PLANNED_DOWN·UNPLANNED_DOWN을 차감해 가동시간과 가용성을 구하고, 성능(사이클타임 효율)·품질(QC 수율)과 결합해 OEE를 산출한다. 계획/비계획 손실을 분리 표기하여 각 라인 고유 시간패턴이 공정하게 반영된다.

SCREEN ID	PP-OEE
PERIOD	Day / Week / Month / custom
SOURCE	PP_LineStateLog (PP-ODM)
TABLES	PP_LineStateLog , PP_Line0EE , MD_LineTimePattern

RBAC	PP.LineOEE
DOWNSTREAM	RPT-06 OEE · RPT-07 KPI

Loading Time = ∑ MD-29 OPERATING minutes | Operating Time = Loading - Planned Down - Unplanned Down

Availability = Operating Time ÷ Loading Time

Performance = (Ideal cycle × Qty) ÷ Operating Time

Quality = Good Qty ÷ Total Qty → OEE = Availability × Performance × Quality

02

Screen Mockup

화면 목업 — OEE 손실 워터폴 + 라인 롤업

● PP-OEE · LINE AVAILABILITY & OEE · WEB PC

A-MES · PP · Line Availability & OEE

PP > Line OEE

Period: 2026-05 (Month) ▼

LINE-INJ-01 ▼

Calculate

Export

82.4%
OEE

91.0%
Availability · 가용성

96.5%
Performance · 성능

93.8%
Quality · 품질

TIME-LOSS WATERFALL · 시간손실 워터폴 (LINE-INJ-01, 2026-05)

28,800
Loading 부하시간

-4,200
Planned Dn 계획비가동

-1,640
Unplanned Dn 실제비가동

22,960
Operating 가동시간

-800
Perf. Loss 성능손실

-1,370
Quality Loss 품질손실

20,790
Valuable 가치가동

Line	Loading min	Planned Dn	Unplanned Dn	Avail.	Perf.	Q
LINE-INJ-01 사출 1 라인	28,800	4,200	1,640	91.0%	96.5%	93
LINE-INJ-	13,500	1,800	2,310	82.9%	94.1%	92

Line	Loading min	Planned Dn	Unplanned Dn	Avail.	Perf.	Q
02 사출 2 라인						
LINE- PNT- 01 도장 1 라인	39,600	3,600	980	95.5%	97.2%	90

03 UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-01	Period Selector 기간 선택	SELECT	Y	Day / Week / Month / custom range. Drives the rollup window.	This Month
CMP-02	Line Selector 라인 선택	SELECT	N	Single line drives the waterfall; All lines drives the rollup table.	All
CMP-03	OEE KPI Tiles OEE KPI	AUTO	—	OEE / Availability / Performance / Quality. Green $\geq$ 85%, amber 70-85%, red $<$ 70%.	—
CMP-04	Time-loss Waterfall 손실 워터폴	AUTO	—	Loading → Planned Dn → Unplanned Dn → Operating → Perf. loss → Quality loss → Valuable. Bars from PP_LineStateLog.	—
CMP-05	Line Rollup Table 라인 롤업 표	AUTO	—	One row per line with the full loss breakdown and OEE; sortable.	OEE desc
CMP-06	Calculate 산출	BUTTON	N	Recomputes the rollup; persists a	—

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
				PP_LineOEE snapshot for the period.	
CMP-07	Export 내보내기	BUTTON	N	Exports to Excel / PDF; feeds RPT- 06 OEE and RPT- 07 monthly KPI.	—

04 Event & Action Flow

이벤트 흐름 · 롤업 산출

1

USER select · period + line

User picks a period and line; the screen queries PP_LineStateLog minutes within the window.

기간·라인 선택 → 윈도우 내 PP_LineStateLog 분(分) 조회.

API GET /api/pp/line-oee?from=&to=&line=

2

SYSTEM aggregate · time-loss model

Sums minutes by state, derives Loading / Operating Time and Availability per line.

상태별 분 합산 → 라인별 부하·가동시간·가용성 도출.

CALC Availability = Operating ÷ Loading

3

SYSTEM join · performance + quality

Joins WO output (ideal cycle vs actual) and QC good-qty to compute Performance and Quality, then OEE.

WO 실적(이상 사이클 대비 실제)·QC 양품수량을 결합해 성능·품질·OEE 산출.

CALC OEE = A × P × Q

4

USER click · [Calculate]

Persists a PP_LineOEE snapshot row per line for the period so the figure is auditable and stable.

기간별 라인 PP_LineOEE 스냅샷 저장 — 수치 감사·고정.

API POST /api/pp/line-oee/snapshot

SCHEDULED nightly · auto-rollup

A nightly job snapshots the prior day's OEE and pushes it to RPT-06 and RPT-07.

야간 배치가 전일 OEE를 스냅샷하여 RPT-06·RPT-07로 전달.

DOWNSTREAM RPT-06 Equipment OEE · RPT-07 Monthly KPI

05

Exception Handling

예외 처리 · 오류 코드

E-OEE-001

Unlogged Downtime in Period

미입력 비가동 포함

CAUSE	Period contains UNPLANNED_DOWN events with no reason logged.
DETECT	On calculate.
ACTION	OEE shown as provisional; banner links to PP-DTL backlog.

⚠ PROVISIONAL OEE

E-OEE-002

No State Log

상태 로그 없음

CAUSE	PP_LineStateLog empty for the selected window (PP-ODM not run).
DETECT	On query.
ACTION	Cannot compute; prompt to verify PP-ODM coverage.

🚫 NO DATA

E-OEE-003

Signal-Loss Minutes

신호두절 구간

CAUSE	UNKNOWN-state minutes (E-ODM-001) inside the period.
DETECT	Aggregate scan.
ACTION	Excluded from Loading; coverage % shown beside OEE.

⚠ PARTIAL COVERAGE

E-OEE-004

OEE Below Target

OEE 목표 미달

CAUSE Line OEE under the 75% target.

DETECT On calculate.

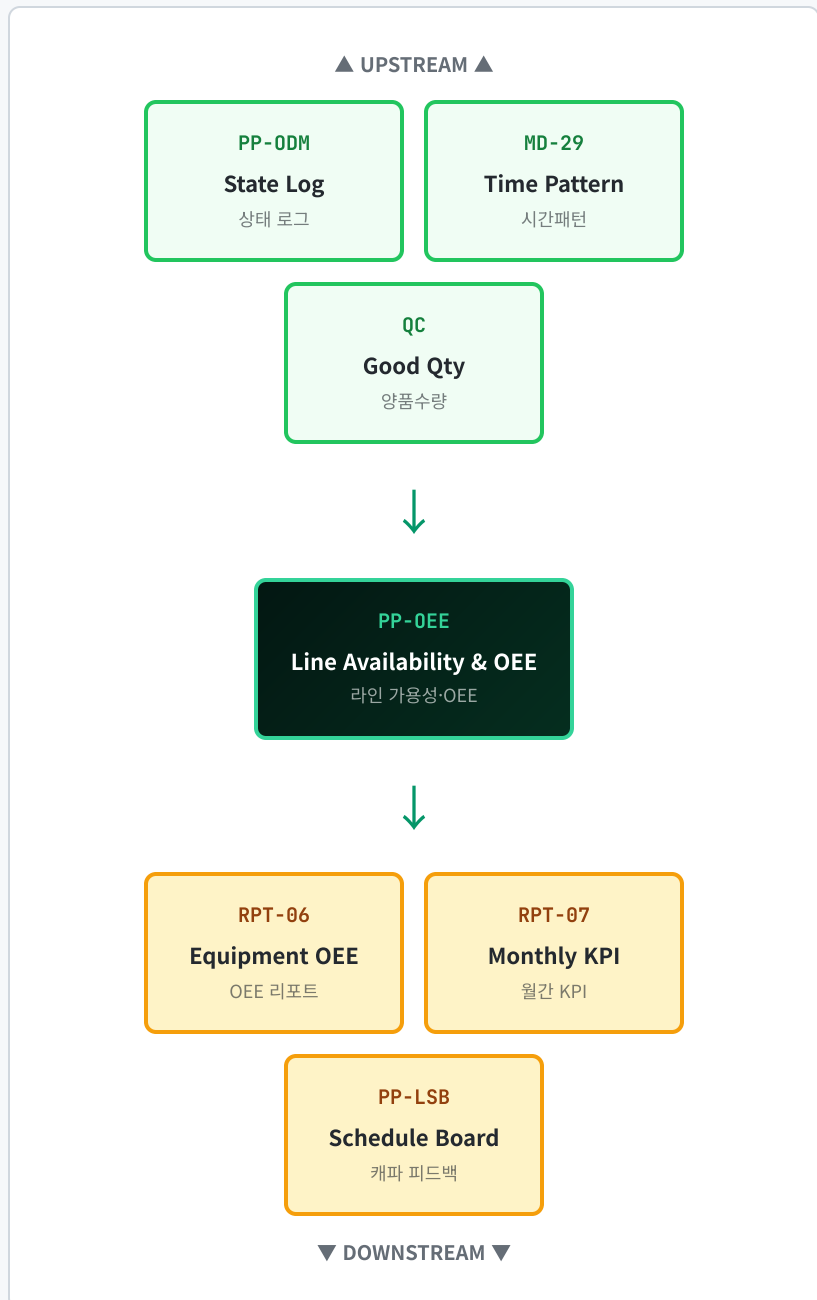
ACTION Row red; alert to PLAN manager for loss review.

⚠ OEE < TARGET

06

Linked Screens & Data Flow

연계 화면 · 상·하류 흐름



RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-040	Loading Time = Σ MD-29 OPERATING minutes for the line and period.	부하시간 = 라인 · 기간의 MD-29 OPERATING 분 합계.	AUT
BR-PP-041	Availability counts only unplanned downtime as loss; planned downtime is excluded.	가용성은 실제 비가동만 손실로 집계 — 계획 비가동 제외.	AUT
BR-PP-042	OEE = Availability $\times$ Performance $\times$ Quality, snapshotted per period.	OEE = 가용성 $\times$ 성능 $\times$ 품질, 기간 별 스냅샷 저장.	AUT
BR-PP-043	A period with unlogged downtime yields a provisional (non-final) OEE.	미입력 비가동 포함 기간은 잠정(비확정) OEE로 표기.	ALT
BR-PP-044	Line OEE below the 75% target alerts the PLAN manager for loss review.	라인 OEE 75% 목표 미달 시 PLAN 관리자 손실 검토 경보.	ALT

A-MESIndex

WH WarehousePP PlanINJ InjectionIMG WrappingPNT PaintingQC QualityFG Finished GoodsMNT Main

L1A-MES Index > L2VOL.4 · PP Production Plan > L3PP-DTL · Downtime Log Entry

VOL.4 · PP MODULE

PRODUCTION PLAN

생산계획

L3 SCREENS · 화면 상세

PP-01Forecast MgmtForecast 관리

PP-02SAP B1 Import공급계획 수신

PP-03Plan & Confirm생산계획 수립

PP-04Work Order Mgmt작업지시 관리

PP-05MRP자재 소요 계획

PP-06Purchase Req.자재발주 요청

PP-07WO ReleaseWO 발행

PP-CALProduction Calendar생산 캘린더

PP-OTDDelivery KPInap기 현황

LINE SCHEDULING · 라인 스케줄링

PP-LSBLine Schedule Board라인 스케줄 보드

PP-ODMOperating/Downtime Monitor가동·비가동 모니터

PP-OEELine Availability & OEE라인 가용성-OEE

PP-DTLDowntime Log Entry비가동 로그 입력

MD-29Line Time Pattern라인 시간패턴

ON THIS PAGE

01 Overview

02 Screen Mockup

03 UI Components

VOL.4 · PP · LINE SCHEDULING · 비가동 로그 입력

PP-DTLDOWNTIME LOG ENTRY

비가동로그입력 · 사유 기록

DEVICEPOP 터치패널 / PDA

PHASE2nd Dev

ROLELine Operator

FREQOn event · 이벤트

URL/pop/downtime-log

01Overview

화면 개요 · 목적 · 컨텍스트

A POP touch-panel screen where the line operator records the **reason for an unplanned downtime event** detected by PP-ODM. The event arrives pre-filled with line, start/end minute and live duration; the operator taps a downtime **reason code** (MD-25, filtered to ReasonType=DOWNTIME) and an optional 4M1E **cause** (MD-22), enters a comment if the reason requires it, and confirms. The closed event with its reason flows to PP-ODM and PP-OEE so availability loss is correctly attributed.

PP-ODM이 감지한 **실제 비가동 이벤트의 사유**를 라인 작업자가 기록하는 POP 터치패널 화면. 이벤트는 라인·시작/종료 분·실시간 경과시간이 미리 채워져 도착하며, 작업자는 비가동 **사유코드**(MD-25, ReasonType=DOWNTIME 필터)와 선택적 4M1E **원인**(MD-22)을 터치 선택하고, 사유가 요구하면 코멘트를 입력한 뒤 확정한다. 사유가 기록된 종료 이벤트는 PP-ODM·PP-OEE로 전달되어 가용성 손실이 정확히 귀속된다.

SCREEN ID	PP-DTL
DEVICE	POP 15" touch · PDA

REASON SRC	MD-25 (DOWNTIME) · MD-22
TABLES	PP_LineDowntimeLog , PP_LineStateLog
RBAC	PP.DowntimeLog
DOWNSTREAM	PP-ODM · PP-OEE · MNT-06

02 Screen Mockup

화면 목업 — POP 터치패널

● PP-DTL · DOWNTIME LOG ENTRY · POP 터치패널

A-MES POP · 비가동 사유 입력

LINE-PNT-01 · 14:37

⚠ UNPLANNED DOWN

LINE-PNT-01 · 도장 1라인

14:24 정지 → 진행중 · 계획상 OPERATING 구간

00:13:24

① 비가동 사유 · REASON (MD-25)



설비 고장

RSN-DT-BRK



자재 대기

RSN-DT-MAT



품질 정지

RSN-DT-QC



금형 트러블

RSN-DT-MLD

② 원인 (4M1E) · CAUSE (MD-22)

MATERIAL

자재

METHOD

방법

MAN

사람

💬 코멘트 (필수) — 분체 로트 교체 지연, 입고 대기 ...

취소

✓ 사유 확정 · 이벤트 종료

터치 1·2단계 선택 → 코멘트 → 확정, 진행중 이벤트는 종료 시 EndMin 자동 확정.

03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

ID	COMPONENT 컴포넌트	TYPE	REQ. 필수	VALIDATION / BEHAVIOR 검증 · 동작	DEFAULT
CMP-01	Event Card 이벤트 카드	AUTO	—	Shows line, start minute, live duration. Open events tick every second until reason confirmed.	—
CMP-02	Reason Tiles 사유 타일	TOUCH	Y	Big touch tiles loaded from MD-25 where ReasonType=DOWNTIME and AppliesToModule covers the line's process.	(none)
CMP-03	Cause Tiles (4M1E) 원인 타일	TOUCH	N	Optional MD-22 cause; tiles filter by the selected reason's process scope.	(none)
CMP-04	Comment Field 코멘트	TEXT	—	Becomes required when the chosen reason has <code>RequiresComment=TRUE</code> ; on-screen keyboard.	(empty)
CMP-05	Confirm Button 확정	BUTTON	Y	Disabled until a reason is picked and any required comment entered. Writes PP_LineDowntimeLog, closes the event.	Disabled
CMP-06	End-time Capture 종료시각	AUTO	—	For an in-progress event, EndMin is set to the equipment RESUME minute or the confirm minute, whichever first.	—
CMP-07	Pending Queue 미입력 대기	LIST	—	Lists all unlogged downtime events for the line so the operator can clear the backlog.	—

04

Event & Action Flow

이벤트 흐름 · 사유 입력 → 종료

1

SYSTEM push · downtime event open

PP-ODM detects an unplanned downtime and pushes an open event to the line's POP panel.

PP-ODM이 실제 비가동을 감지하여 라인 POP 패널에 진행중 이벤트를 푸시.

PUSH WS event: `downtime.open {lineId, startMin}`

2

USER tap · reason + cause

Operator taps a DOWNTIME reason tile (MD-25) and, if needed, a 4M1E cause tile (MD-22).

작업자가 DOWNTIME 사유 타일(MD-25)과 필요 시 4M1E 원인 타일(MD-22)을 터치.

STATE reason set · cause optional

3

SYSTEM validate · comment requirement

If the chosen reason has RequiresComment=TRUE, the comment field is enforced before Confirm enables.

선택 사유가 코멘트 필수면 Confirm 활성화 전 코멘트 입력 강제.

RULE BR-PP-046 · MD_ReasonCode.RequiresComment

4

USER tap · [사유 확정]

Writes a PP_LineDowntimeLog row (line, start/end minute, reason, cause, comment, operator) and closes the event.

PP_LineDowntimeLog 행 기록(라인 · 시작/종료분 · 사유 · 원인 · 코멘트 · 작업자) → 이벤트 종료.

API `POST /api/pp/downtime-log`

5

SYSTEM complete · attribute loss

The reasoned downtime updates PP-ODM (event cleared), feeds PP-OEE availability loss and mirrors to MNT-06 if reason is a breakdown.

사유 기록된 비가동이 PP-ODM 갱신(이벤트 해소)·PP-OEE 가용성 손실 반영 · 고장 사유면 MNT-06 동기화.

DOWNSTREAM PP-ODM clear · PP-OEE recalc · MNT-06 downtime

05

Exception Handling

예외 처리 · 오류 코드

E-DTL-001

Confirm Without Reason

사유 미선택 확정

CAUSE	Operator taps Confirm with no reason selected.
DETECT	Confirm guard.
ACTION	Confirm stays disabled; reason tiles pulse.

REASON REQUIRED

E-DTL-002

Comment Missing

코멘트 누락

CAUSE	Reason requires a comment but field is empty.
DETECT	On confirm.
ACTION	Comment field highlighted; keyboard opens.

COMMENT REQUIRED

E-DTL-003

Event Auto-Resumed

자동 재가동

CAUSE	Equipment resumes while operator is still picking a reason.
DETECT	Signal RESUME.
ACTION	EndMin frozen at resume; reason entry continues for the closed span.

AUTO-CLOSED · LOG STILL DUE

E-DTL-004

Offline POP

POP 오프라인

CAUSE	Network drop while confirming.
DETECT	API timeout.
ACTION	Log queued locally, auto-synced on reconnect.

OFFLINE · QUEUED

06

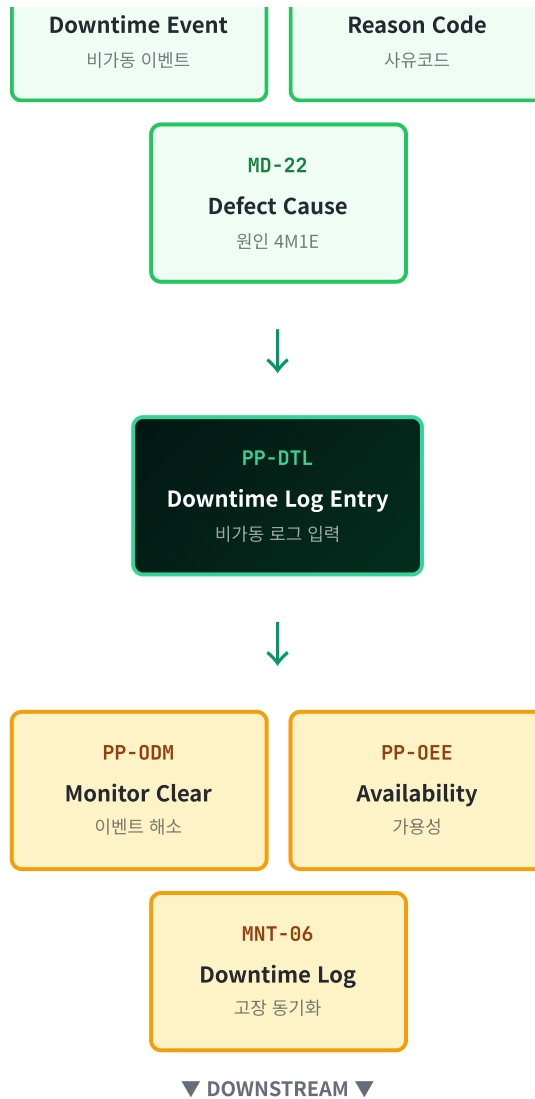
Linked Screens & Data Flow

연계 화면 · 상·하류 흐름

▲ UPSTREAM ▲

PP-ODM

MD-25



07 Applicable Business Rules

적용 업무 규칙

RULE ID	RULE (EN)	규칙 (KO)	ACTION
BR-PP-045	Every unplanned downtime event must be closed with a DOWNTIME reason code.	모든 실제 비가동 이벤트는 DOWNTIME 사유코드로 종료해야 함.	BLK
BR-PP-046	A reason with RequiresComment=TRUE blocks confirm until a comment is entered.	코멘트 필수 사유는 코멘트 입력 전 확정 차단.	BLK
BR-PP-047	An in-progress event's EndMin is fixed at the	진행중 이벤트의 종료 분은	AUT

RULE ID	RULE (EN)	규칙 (KO)	ACTION
	equipment resume minute.	설비 재가동 시각으로 확정.	
BR-PP-048	A breakdown-type reason mirrors the event to the MNT-06 downtime log.	고장 유형 사유는 이벤트를 MNT-06 비가동 로그에 동기화.	AUT
BR-PP-049	Offline POP entries are queued locally and auto-synced on reconnect.	오프라인 POP 입력은 로컬 큐 저장 후 재접속 시 자동 동기화.	AUT